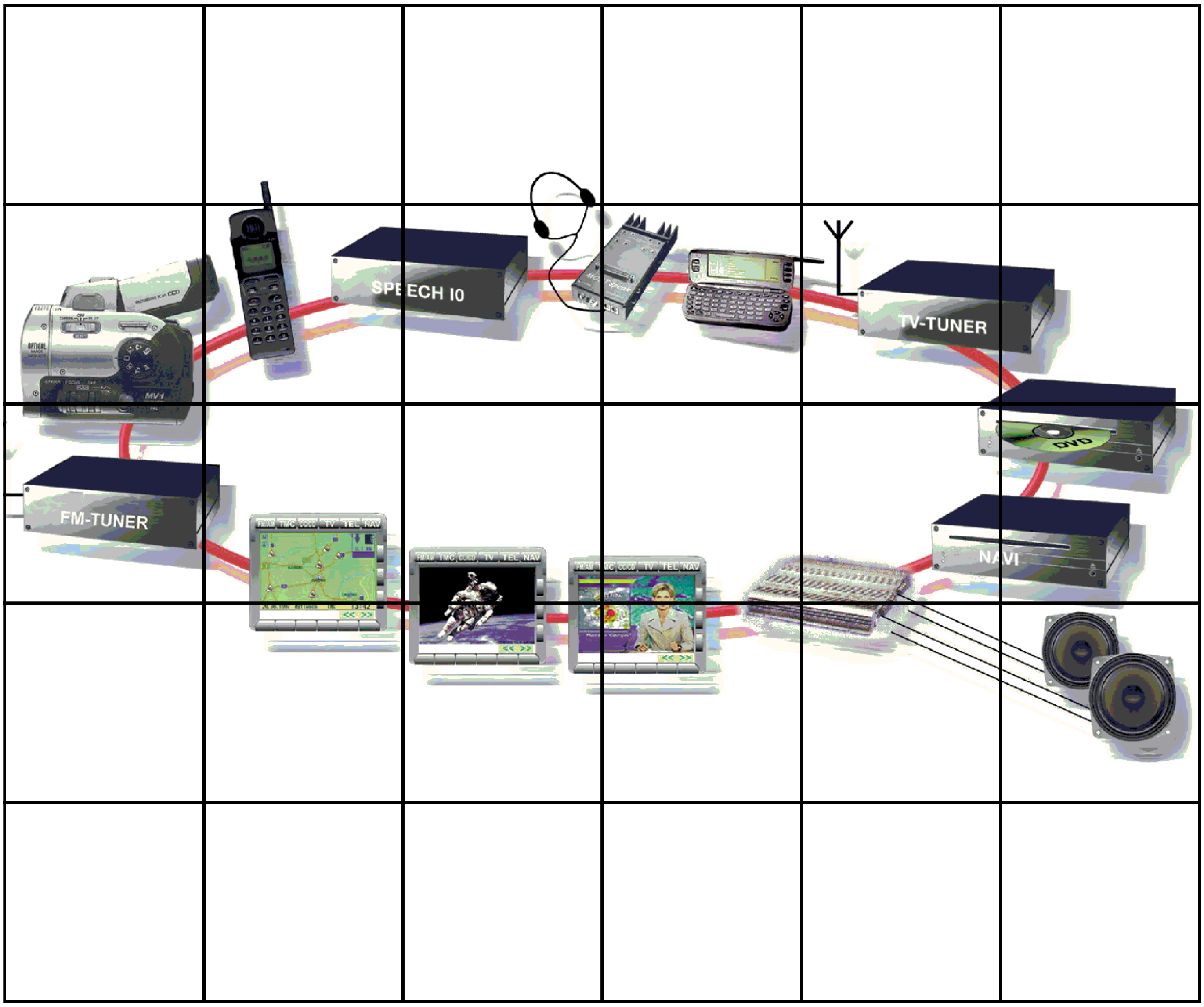
BMW

Service Training

Information/Communication

Seminar Working Material



NOTE
The information contained in this training course manual is intended solely for participants of the BMW Service Training course.
Refer to the relevant "Technical Service" information for any changes/ supplements to the Technical Data.

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Contents

Introduction

What was defined in the E38 as a comprehensive networking of the control units has been further extended in the E65. To satisfy increasing customer requests for more ride comfort, e.g. a high-quality audio system, navigation, telephone etc., a faster and more flexible audio communication assembly is provided.

The digital networking of the audio communication assembly is achieved by means of the MOST (Multi Oriented Systems Transport) network.

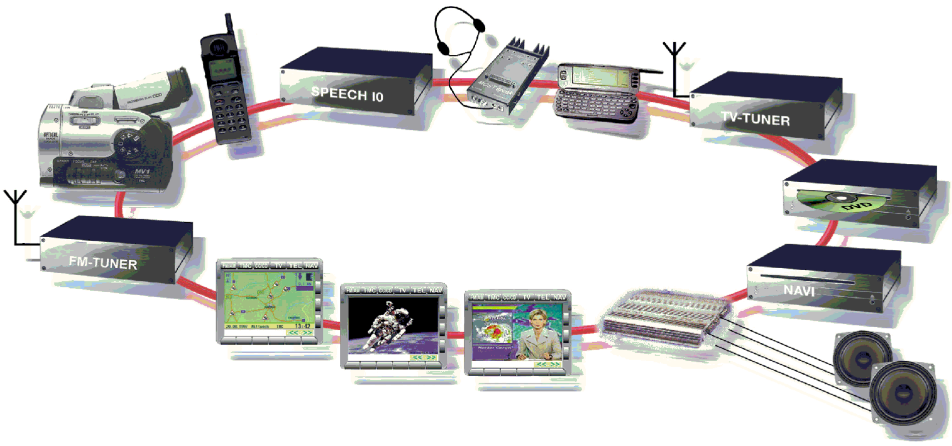

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Fig. 1: Overview of possible multimedia applications in the future

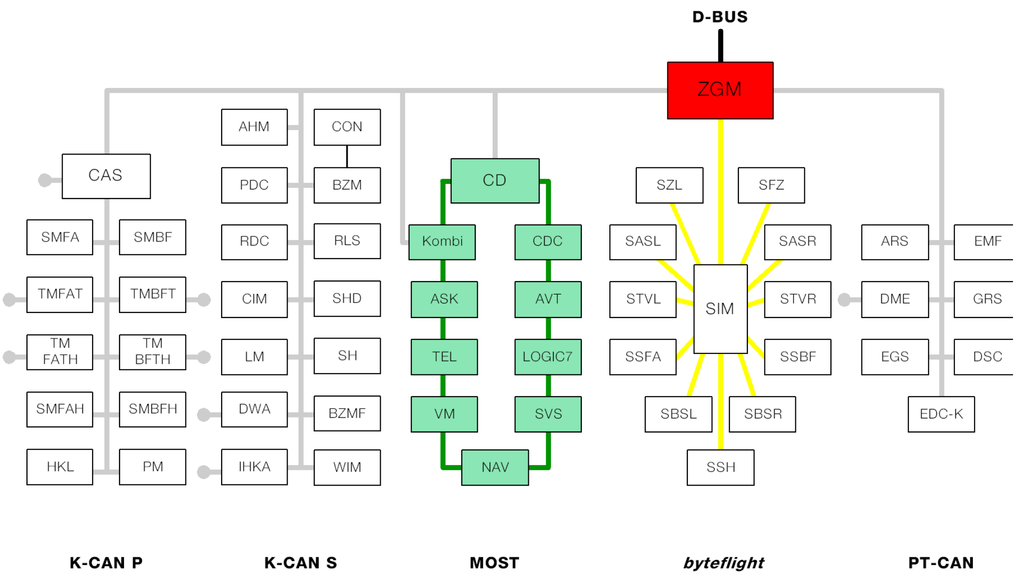

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Fig. 2: Overview of the MOST applications in the E65

Index	Description
CD	Control Display
CDC	Audio system CD changer
AVT	Aerial amplifier/tuner
LOGIC 7	Amplifier for TOP-HiFi
SVS	Voice processing system
NAV	Navigation system
VM	Video module
TEL	Telephone module
ASK	Audio System Controller
Kombi	Instrument cluster

- System philosophy

With the new concept of the audio communication network the way of achieving a comprehensive networking in the E38 is therefore extended further in the E65.

The current philosophy of single systems simply connected to each other to exchange data or functions has not been maintained in the E65.

The most important features of the audio communication assembly of the E65 are:

● Central controls and display with a Control Display

Via the centralised and unified operation, a unified design of all the functions is achieved in the display.

● Central audio management in the audio system controller

All audio signals are collected in the ASK processed and distributed (via the loudspeakers). This creates high-quality overall acoustics. The voice and signal outputs are integrated.

● New audio concept

With an optimal position of the loudspeakers, you can reduce their number without altering the global acoustic effects.

- Voice operation

The system offers a verbal handling of on-board features.

- Equivalent rear equipment

The whole extent of the on-board features could be provided in an equivalent way in the rear compartment (future).

- Multimedia networking of the control unit

Control and multimedia data, video data excepted, are conveyed on a common bus.

- Integrated system control

A unified control concept, which must be applied to all control devices of the network, takes into account and enables comprehensive information links between the individual control modules.

- New customer functions

From the single functions in the separate control modules it is possible to create new general features, highly effective for the customers via the comprehensive interlinks.

- Plug & Play of MOST users (later)

By defining of standard interfaces you obtain a significant reduction in the number of variants. The expansion of the MOST network is possible with no major effort.

MOST bus

- General

MOST is a communications technology for multimedia applications, specially developed for use in the automobile.

MOST stands for "Multimedia Oriented System Transport." The MOST bus uses light to transmit data.

- System overview

MOST technology

Until very recently, only very few entertainment-related control units were networked. In the course of development, the number of components rose continuously.

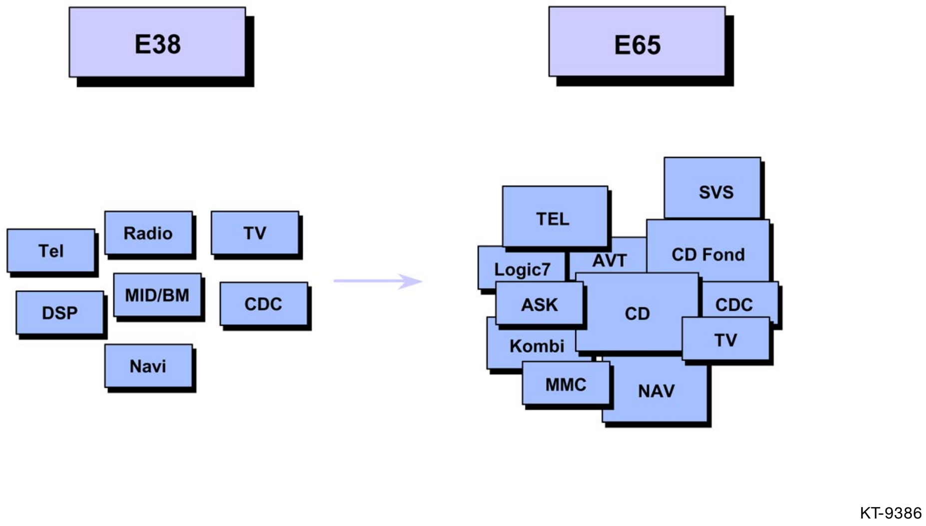


Fig. 3: Development in the area of entertainment

Index	Description	Index	Description
TEL	Telephone	LOGIC7	Top HiFi amplifier
DSP	Digital Sound Processing	ASK	Audio System Controller
Radio	Radio	KOMBI	Instrument cluster
MID/BM	Multi-information display/ On-board monitor	AVT	Aerial amplifier and tuner

Index	Description	Index	Description
NAVI	Navigation system	CD	Control Display
TV	TV	SVS	Voice processing system
CDC	Compact Disc Changer, CD changer	MMC	Multimedia changer

In addition, the scope of functions of individual components has been extended considerably.

In particular, however, completely new logic networking means that all the components are gradually becoming one system. Individual functions work together and produce a high-quality overall system. This results in significant growth in system complexity.

This new dimension of system complexity can no longer be managed using the existing bus systems.

MOST multimedia network

MOST technology meets 2 essential requirements:

1. The MOST bus transports control data as well as data from audio, video, navigation and other services (SMS = Short Message Service, TMC = Traffic Message Channel).
2. MOST technology provides a logical framework model for control of the variety and complexity of data: the MOST Application Framework. The MOST Application Framework organises the functions of the overall system.

MOST is able to control and dynamically manage functions that are distributed in the vehicle.

Principle of a multimedia network

An important feature of a multimedia network is that it transports not only control data and sensor data, e.g. the CAN bus and I bus (instrumentation bus).

A multimedia network can also carry digital audio and video signals and graphics as well as other data services.

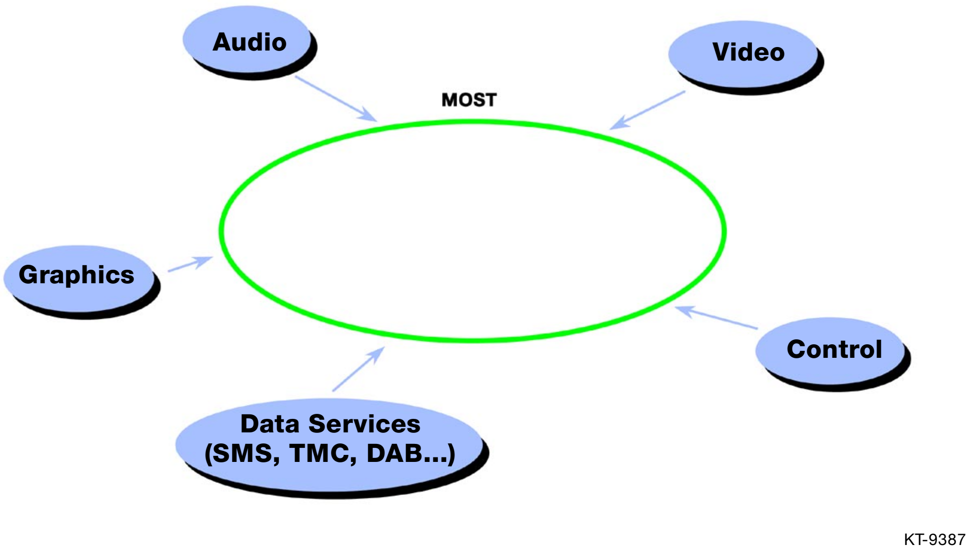


Fig. 4: Principle of a multimedia network

Data volumes

The aim is that in the near future all occupants can start different services simultaneously, e.g.:

- The driver calls up navigation information
- The passenger watches TV
- A rear seat passenger listens to a CD, and
- The other rear seat passenger watches a video

The data volumes this requires produce the following picture:

Application	Band width	Data	Data format
AM-FM CC Check-Control CD Audio MD Telephone SVS	1.4 Mbit/s	1 stereo channel	synchronous
TV CD Video	1.4 Mbit/s 1.4 Mbit/s	audio MPEG 1 Video	synchronous
DVD	2.8-11 Mbit/s	MPEG 2 Video	synchronous and asynchronous
Navigation	250 kbit/s 1.4 Mbit/s 1.4 Mbit/s	vector data MPEG 1 Video voice output	asynchronous synchronous synchronous
Telematics services	a few byte/s ...		asynchronous

Using MOST, there is already the capability today to transport these large data volumes.

- Functional description

Data transport

MOST currently has a bandwidth of 22.5 Mbit/s (as at May 2001).

In order to meet the different requirements of the applications regarding data transport, each MOST message is divided into three parts:

- Control data
- Asynchronous data: e.g. navigation system, arrow representation
- Synchronous data: e.g. audio, TV, video signals

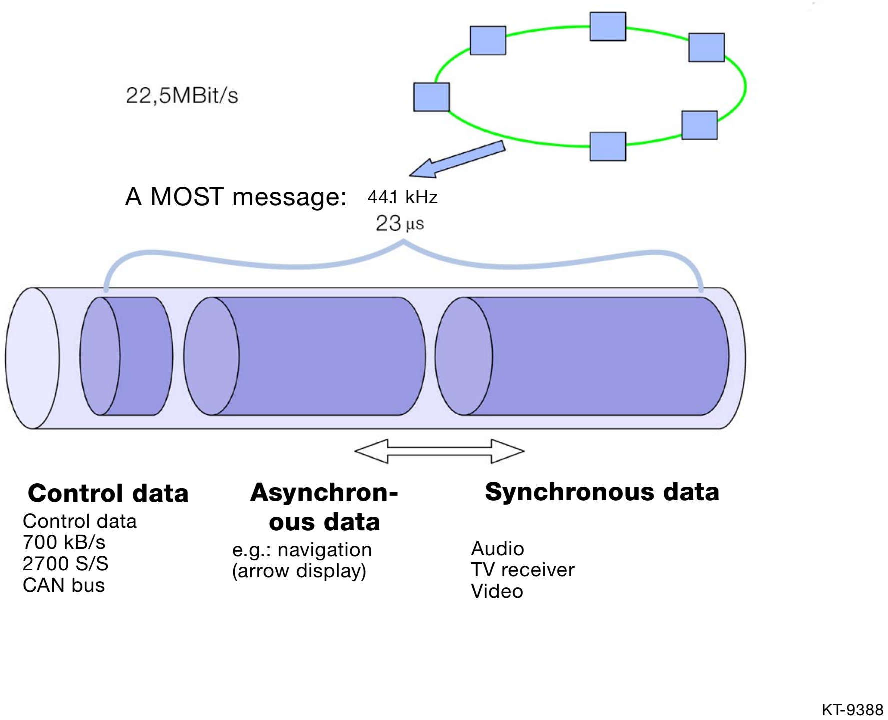


Fig. 5: Data transport on the MOST bus

The control channel controls the functions and devices in the network. The control channel can be compared to the CAN bus. The control channel has a bandwidth of 700 kbit/s. This corresponds to around 2700 messages per second.

For the data transmission of synchronous and/or asynchronous data, there is a total of 60 bytes. The limit is variable, e.g. 20 bytes of synchronous data and 40 bytes of asynchronous data.

Optical waveguide transmitter

A driver is fitted in the transmitter. The driver energises an LED (light-emitting diode). The LED transmits light signals on the MOST bus (650 nm light, i.e. red, visible light).
The repeat frequency is 44.1 MHz.

The sensing frequency on a CD player and for audio is 44.1 MHz; this means than no additional buffer is required - another reason why this bus system is so efficient.

Optical waveguide receiver

The receiver receives the data from the MOST bus.

The receiver consists of

- a diode
- a pre-amplifier
- a wake-up circuit
- an interface that converts the optical signal into an electrical signal

The receiver contains a diode that converts the optical signal into an electrical signal. This signal is amplified and further processed at the MOST network interface.

Optical waveguides

The MOST bus is a plastic optical waveguide.
The MOST bus is coded in green in the E65.
The light wavelength is 650 nm (red light).

The basis of optical waveguide technology is described in detail in the background chapter "Optical waveguides."

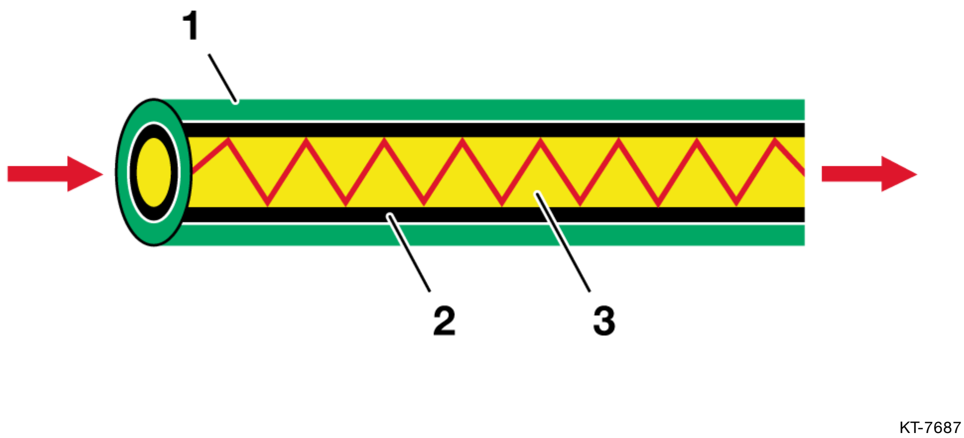


Fig. 6: Cross-section through the optical waveguide

Index	Description	Index	Description
1	Padding sleeve	3	Fibre core
2	Cladding		

The MOST bus requires the following converter components:

- Optical waveguide transmitter
- Optical waveguide receiver

Each control unit of the MOST framework contains a transmitter and a receiver.

The transmitter and receiver have been developed by BMW. The low quiescent current properties of the transmitter and receiver enable optical wake-up by the MOST bus.

Control unit/control unit connection

The MOST ring is composed of optical point-to-point connections between 2 control units.

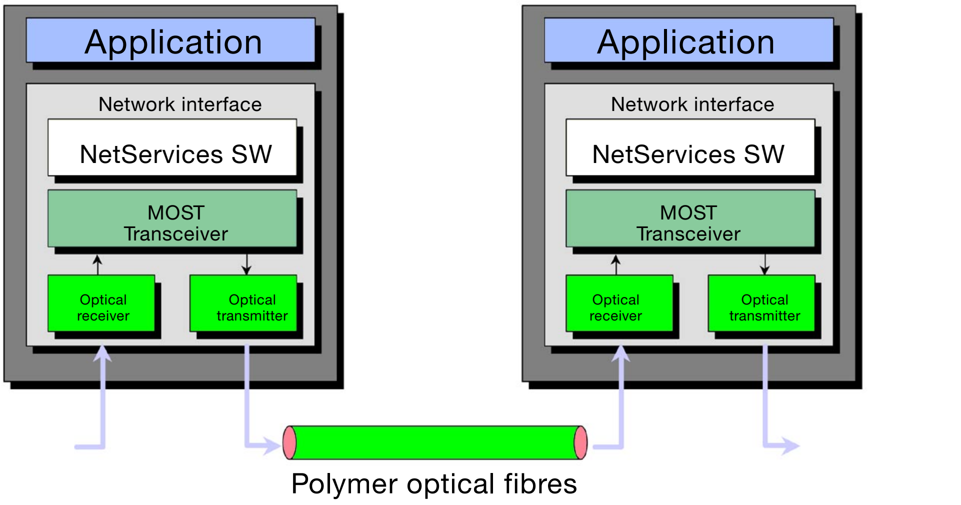


Fig. 7: Control unit/control unit connection

Each control unit has a network interface. The network interface consists of

- an opto-electrical converter (optical waveguide receiver, already mentioned)
- an opto-electrical converter (optical waveguide transmitter, already mentioned)
- a MOST transceiver (interface between the optical waveguide receiver/transmitter and the electronic network driver)
- a network driver, the so-called NetServices.

MOST as a control system for distributed functions

In addition to data transport, the second essential task of the MOST technology is to control distributed functions: "The MOST Application Framework."

From the traditional vehicle electrical system to the "MOST Application Framework"

The starting point is a traditional vehicle electrical system configuration:

- A number of control units are networked via a bus. They are regarded as black boxes. At best, only the name of the box reveals the functional content (e.g. telephone). The individual boxes interchange signals.
- There is no higher level control unit that specifies certain rules.
- There are only fixed rules concerning the message structure.
- A hierarchy of what controls what is not apparent. Cross influences that are hard to control can occur.

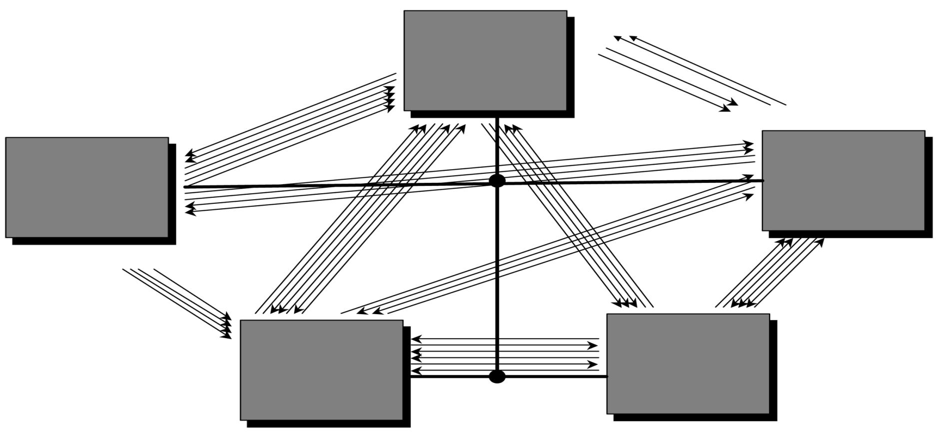


Fig. 8: Traditional vehicle electrical system configuration

This situation is comparable to that of a company without a hierarchy, whereby the functions correspond to the employees. As long as the company remains small, everything works well. When the company becomes larger, however, different structures are required.

The entertainment framework (MOST network, information/ entertainment framework) in vehicles has become very large in the meantime. The entertainment framework needs hierarchical structures and rules for communication.

These hierarchical structures and rules are introduced by the "MOST Application Framework."

In order to throw light on the path from the traditional vehicle electrical system to the "MOST Application Framework," this path is shown here in 3 steps:

1. Model building
2. Application of model building to the control units
3. Implementation of a hierarchical structure

1. Building a model

The first step to achieve order is building a model:

The content of the control units is divided. Each control unit consists of one application. The application is subdivided into function blocks.

For example, a radio contains the function blocks

- tuner
- amplifier, and
- deck

Each function block in turn contains functions. A function is a feature that can be experienced and influenced by the customer. The function block "tuner" contains, for example, the functions

- frequency
- search
- TMC data (Traffic Message Channel) etc.

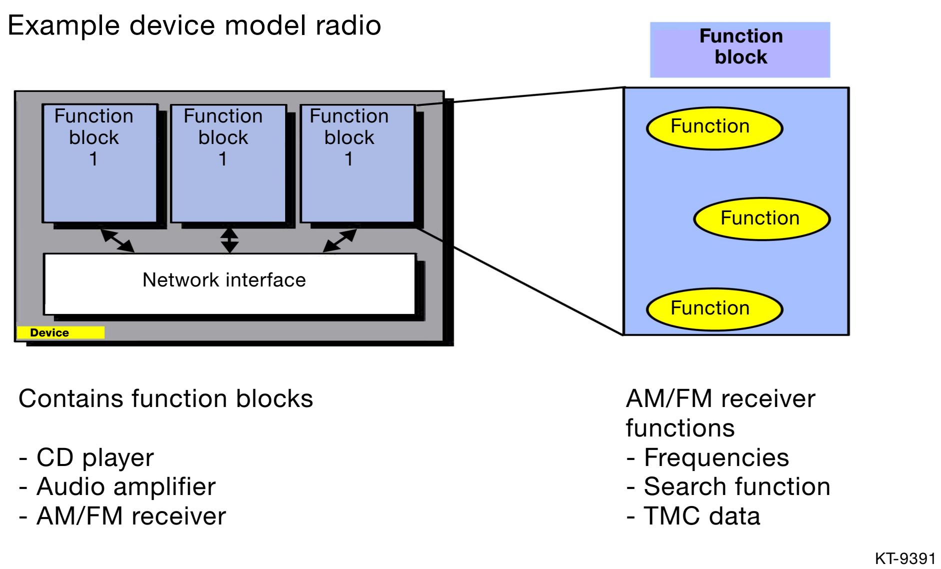


Fig. 9: Building a model: from control unit to function

2. Application of model building to the control units

Model building is applied to each control unit of the network. The black boxes begin to take shape.

As a matter of fact it is unimportant which function is located in which box.

The advantage of this physically independent arrangement lies in the fact that functions can be placed where it makes the most sense to do so (independent of a specific control unit).

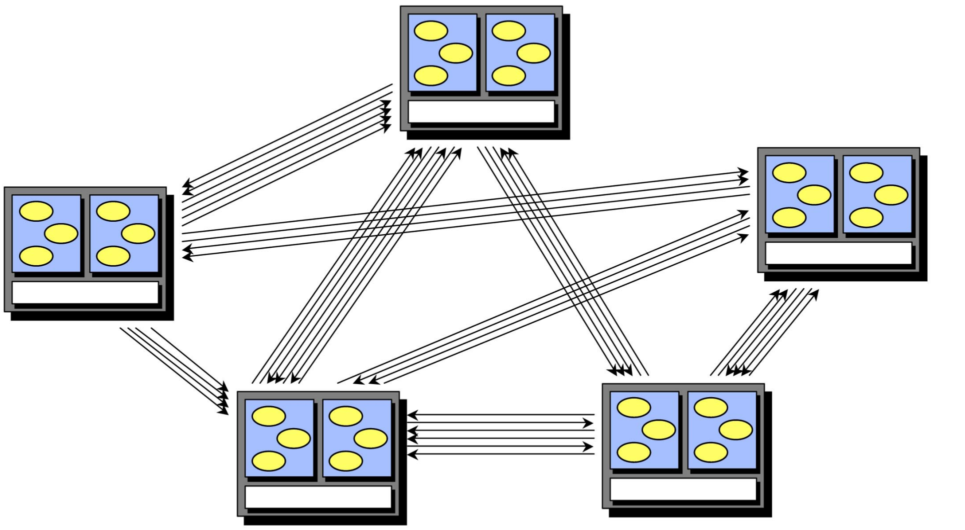


Fig. 10: Function blocks distributed to a number of control units

Let's take the function example of a radio tuner. The radio tuner is located at the rear left in the aerial amplifier (AVT aerial amplifier with tuner). The FM (frequency modulation) signal received by the rear window does not have to be transmitted across a long cable to the front to the radio tuner (which naturally means loss of signal quality). The signal is converted directly in the AVT into a digital signal and sent via the MOST to the Audio System Controller.

Actually, it is of no further interest which function block is located in which box, or which network the boxes are connected to, or how the signals are transmitted.

3. Implementing a hierarchical structure

The last step is implementing a hierarchical structure. The structure consists of 3 levels:

1. System master: master control unit Control Display
2. Controllers: subordinate control units, e.g. ASK Audio System Controller
3. Slaves: functions, e.g. FM tuner

Functions of the slaves

Function units (control units or components) as "slaves:"
Slaves

- offer their function in the system
- do not know who controls them
- do not know in which system they are operated
- have no system knowledge
- have no direct logical link with other function units in the system
- are "Lego blocks" of an entertainment system

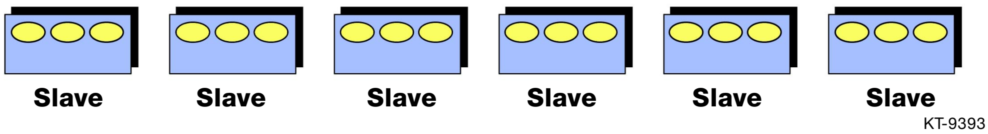


Fig. 11: Function units as slaves

Examples of "slave" function units are

- tuner
- player devices (CD changer)
- amplifiers
- voice processing system
- microphone
- telephone

Functions of the controllers

Function units (control units or components) as "controllers:"

- Control other function units and
- Are controlled by other function units
- Are experts in certain complex functions and
- Control a part of the system and/or
- Represent the complex scope of the system simply up top (to the system master)

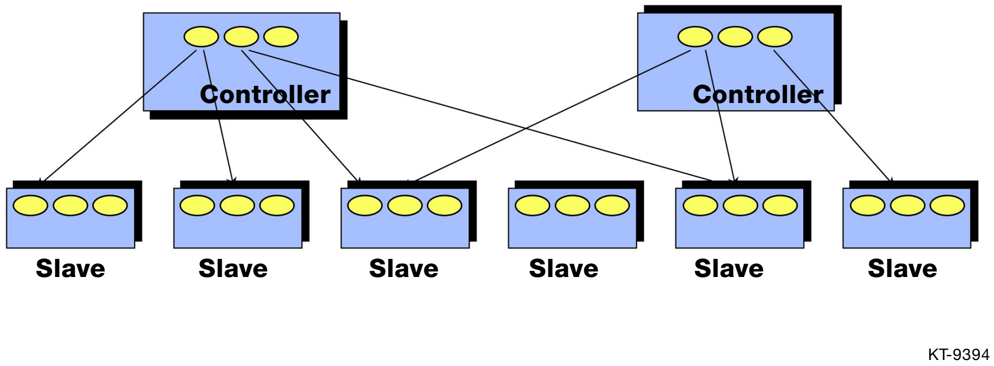


Fig. 12: Function units as controllers

Function example: Audio System Controller:

The "hands-free mode" function is a function of the Connection Master (the Connection Master is integrated in the Audio System Controller).

To set up the "hands-free mode" function, the Connection Master in the ASK must

- interconnect the microphone, telephone and amplifier
- set the mixer
- set the volumes
- etc.

However, this complexity does not appear up top:
The system master Control Display

- does not request these ASK functions
- does not control these ASK functions

Functions of the system master Control Display

The top level of the MOST hierarchy is the system master. The Control Display is the system master. The Control Display

- controls the overall system using the controllers at a high abstract level
- displays the overall system to the user
- directly accesses the slaves, where there is no controller
- has overall knowledge
- links individual function units and function areas to form a system

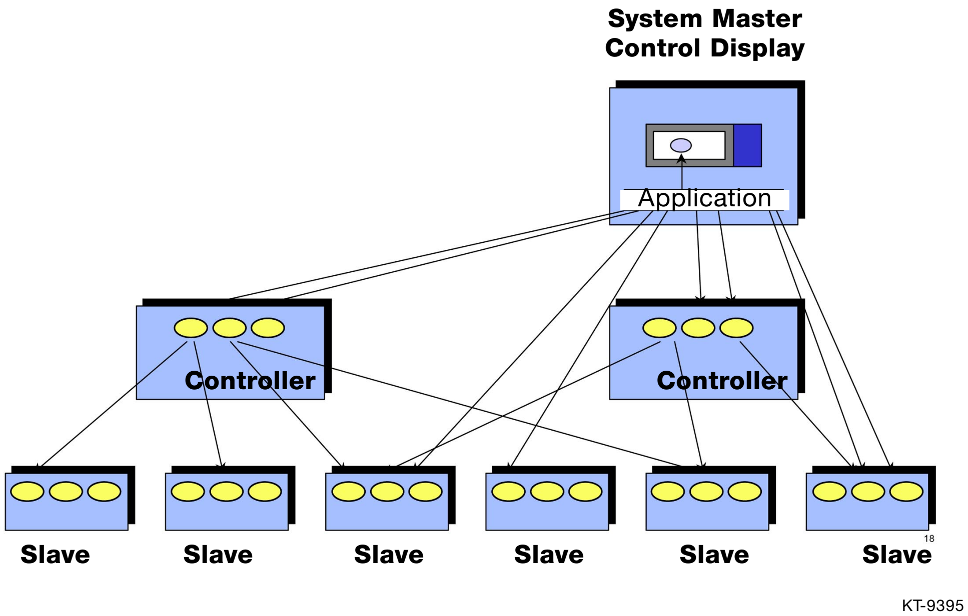


Fig. 13: Control Display as system master

A function example is the "Activate hands-free mode" function during a telephone call with the handset.

Initial situation: the call is in progress. The driver is speaking on the telephone using the handset. The driver presses the FSPOnOFF softkey on the handset to "on" (FSP = hands-free mode).

The following operations are performed:

- The telephone internally sets the property "FSP on" and transmits the information to the Control Display. (The Control Display is the central system master.)
- The Control Display transmits the message to the Audio System Controller (ASK).
- The ASK switches any sound sources that might be playing (CD-player, radio) to "mute."
- The ASK sets up the audio connections necessary for hands-free mode. To do so, the ASK informs the telephone about where the audio data is to be positioned. The telephone confirms that the data is positioned in this way.
- The ASK sends the message to the audio channels: connect loudspeakers and hands-free microphone.
- The ASK transmits the message: activate audio sources.
- As its last action, the ASK sends another message to the audio amplifier: cancel mute. The "hands-free mode" function is now active.
- The ASK transmits as confirmation to the Control Display the message: FSP active.

Advantages of the system

The advantages of this system are summarised here once again:

- High bandwidth.
- The rapidly growing complexity of entertainment services can be better mastered.
- The systems are easy to expand, update and maintain.
- The functions can be freely positioned or placed where it makes most sense to do so. For each vehicle or for each model, a different location could be better.
- The system is already configured for "Plug and Play:" in future, the system will be able to recognise a new device automatically and integrate it in the system network. This makes the replacement and retrofitting of devices simple and problem-free.

- Diagnosis

If faults occur, corresponding fault memory entries can be read out.

Fault memory for MOST

- receiver has not accepted a message (Error_NAK)
- ring break diagnosis carried out (Error_Ring_Diagnosis)
- requesting control unit receives no reply although the relevant control unit is present (Error_Device_No_Answer)

Service information: only repair MOST bus between 2 control units once

The MOST bus may only be repaired once between 2 control units, otherwise losses can become too great.

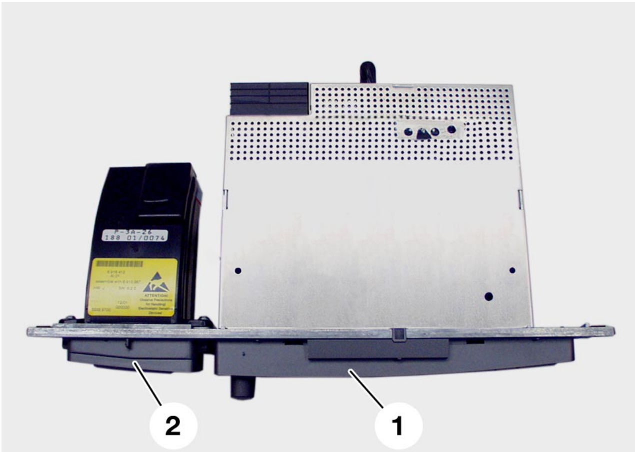
Service information: only repair MOST bus using special tool!

The MOST bus may only be repaired using the prescribed special tool (crimping pliers).

Audio System Controller

- General

For the first time on the E65 a control unit has been developed to control, co-ordinate and output acoustic signals according to a priority. The Audio System Controller, in short **ASK**.



1. ASK

2. Phone Board

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Fig. 14: Audio System Controller and phone board

The ASK Audio System Controller has three main functions in the MOST network.

- Network master
- Audio master
- Connection master

- Network master

The ASK is the Network Master for the MOST. The functions of the Network master are the following:

- **Wake-up, initialisation, power-down**
- **Configuration control**
- **Control of the network operation**
- **Defect code memory**

- Audio Master

As the Audio Master, the ASK must collect, process and distribute all audio signals in the vehicle to the loudspeakers.

The ASK controls all acoustic requests from the Control Display.

Audio data

All audio data from any control unit are converted by the ASK into digital audio NF format at a sampling rate of 44.1 KHz.

Generating acoustic signs

These are acoustic alarm signals, which provide the driver a means of assigning an acoustic signal to a particular system. The different acoustic signals, requested by the various control units, e.g. gongs, PDC, etc., must be generated only in association with an optical display. Thus the request of acoustic signals is made only by control units which are responsible for the optical output of the displays. This includes the Instrument Cluster and the Control Display.

The following acoustic signals can be generated in ASK.

- Signal for the PDC
- Different gongs, which can be altered via various commands.

Examples of audio output

Example: the radio

The radio plays in the background, the PDC is active and a navigation message is output. If a gong request arrives, the radio is shut off and the gong is generated.

Example: the telephone

When you activate the hands-free mode function, all outputs from the loudspeaker are suppressed. Gongs are mixed on demand according to their priority. The voice processing system is not active when is the hands-free mode function is used.
A traffic announcement cannot be output when the hands-free mode function is being used.

Example: voice processing system

When you activate the voice processing system, all outputs from the loudspeaker are suppressed. Gongs are mixed on demand according to their priority. The hands-free mode operation and the navigation messages interrupt the voice processing system.
A traffic announcement cannot be output when the voice processing system is active.

Example: traffic messages

During a traffic announcement, any entertainment source (CD, CC, MD, etc.) stops playing.

Distribution of audio signals

The audio signals are distributed to the following output ports.

Signal	Output port
Mandatory acoustic signs	Front left and/or right
Radio set 1	Front left
Radio set 2	Front right
Two-way intercom system	Front
PDC	Front, front left, front right Rear, rear left, rear right
Telephone (hands-free conversation)	Front left and/or right
Gong with gong priority 1	Front left and/or right
Navigation message	Front left and/or right
Gong with gong priority 2	Front left and/or right
Voice output with SVS	Front left and/or right
Traffic message	Front left and/or right
Gong with gong priority 3	Front left and/or right
Entertainment source	All vehicle loudspeakers

- Connection master

As the connection master, the ASK must provide channels to the equipment connected to the bus and distribute the audio signals on the outputs (loudspeakers).

The connection master also controls the HiFi or the LOGIC 7 HiFi amplifiers.

- ASK variants

The basic or high-performance ASK is mounted together with an audio player in a DIN housing on the central console in place of the previous radio set. The ASK includes four bridge end stages for loudspeaker outputs. The basic ASK also includes the two audio outputs for the HiFi amplifier and the control cable. To satisfy the various user needs, the ASK is offered with the following players.

- ASK with CC player
- ASK with CD player
- ASK with MD player

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Fig. 15: Variants of the Audio System Controller

ASK control elements

The ASK offers the following control elements.

- Rotary pushbutton
- Seek tuning rocker switch
- Eject button

The rotary pushbutton provides the following functions.
The pushbutton is used to switch on and off the audio entertainment system.

The rotary button is used to adjust the audio volume.

With the seek tuning rocker switch you can control several functions in the audio assembly. The assignment is made via the Control Display.

The eject button controls the ejection of the CC, CD or MD.

Audio amplifier HiFi/LOGIC7

- General

In the E65, the audio amplifier is controlled by the ASK.

In the past few years, the demand for high quality in the audio system in vehicles has greatly increased. The functions and features of good home appliances such as Dolby Surround are now expected.

The audio concept of the E65 differs from previous concepts through a reduction to HiFi and LOGIC 7-HiFi (Top-HiFi).

- HiFi amplifier

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Fig. 16: View of the HiFi amplifier

A HiFi system is offered as standard equipment and features a vehicle-specific tuning for the interior (equalising) with a balanced acoustics.

The basic signals delivered by the ASK are amplified in the HiFi amplifier equipped with bridge end stages. The output power is 40 W per channel. The HiFi amplifier is controlled by the ASK.

The medium-range loudspeakers and the tweeters are directly controlled by the ASK.

The HiFi amplifier has no diagnosis capability.

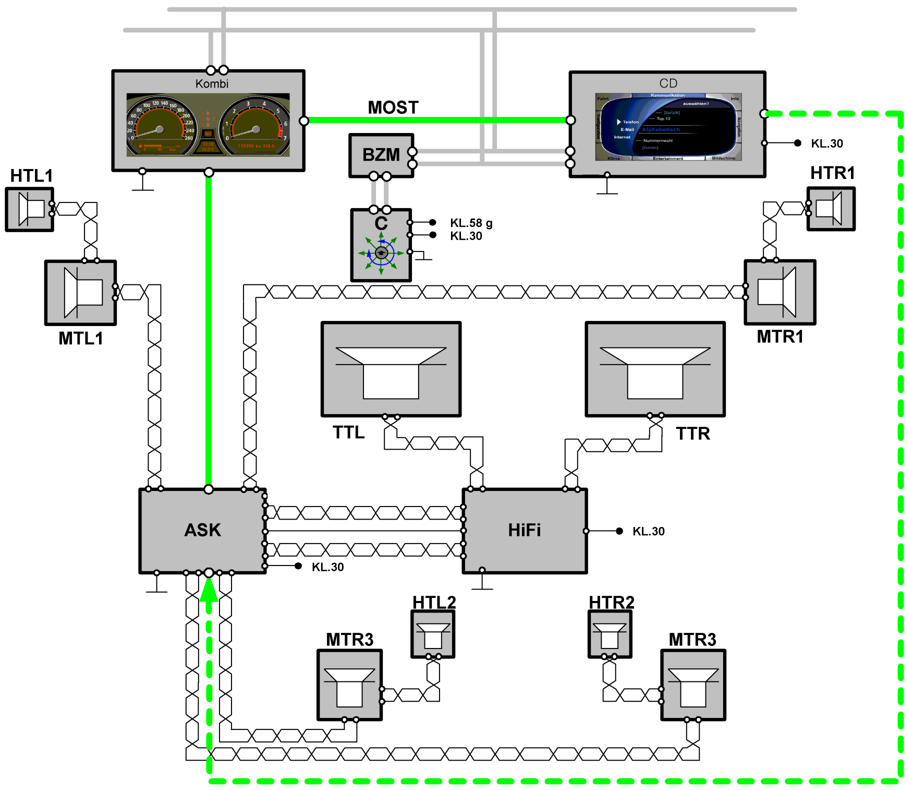

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Fig. 17: Audio HiFi system

Index	Description	HiFi (standard)	LOGIC 7-HiFi (SA)
	The MOST assembly includes other control modules as special equipment.		
Kombi	Instrument cluster	X	X
CD	Control Display	X	X
ASK	Audio System Controller	X	X
HiFi	HiFi amplifier	X	
LOGIC 7 HiFi	TOP HiFi amplifier		X
BZM	Control panel module, centre console	X	X
CON	Controller	X	X
MTL 1	Mid-range speaker for LH front door	X	X
HTL	Tweeter for LH front door	X	X
CS	Central front mid-range speaker		X
MTR 1	Mid-range speaker for RH front door	X	X
HTR	Tweeter for RH front door	X	X

MTL 2	Mid-range speaker for LH rear door		X
MTL 2	Mid-range speaker for RH rear door		X
MTL 3	Mid-range speaker for LH rear hat shelf	X	X
MTL 3	Mid-range speaker for RH rear hat shelf	X	X
HTL 2	Tweeter for LH rear hat shelf	X	X
HTR 2	Tweeter for RH rear hat shelf	X	X
TTL	Front LH woofer	X	X
TTR	Front RH woofer	X	X
MOST	Media Orientated System Transport	X	X
K-CAN S	Bodyshell CAN system	X	X
KL.30	Voltage supply	X	X
KL.58	Lights	X	X

- LOGIC 7 HiFi amplifier

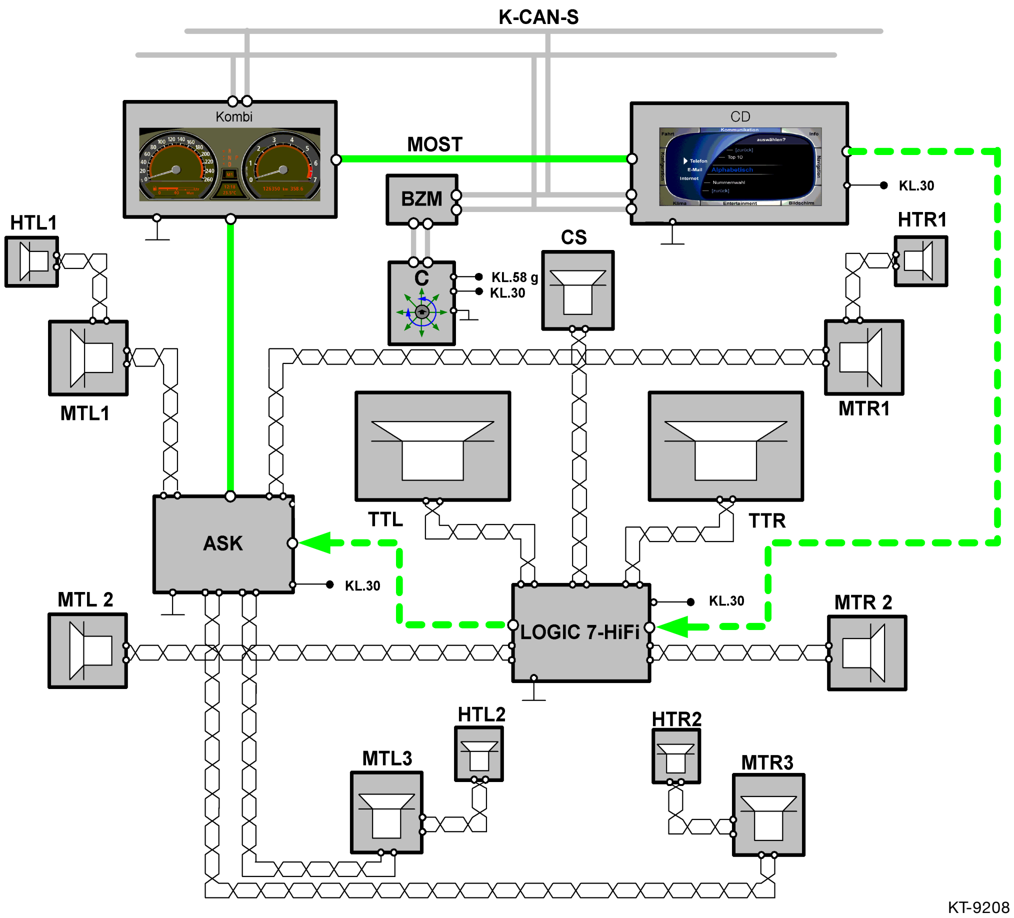


Fig. 18: Audio LOGIC 7 HiFi system

Component and functional descriptions

The LOGIC 7 HiFi amplifier includes a MOST bus connection to convey all audio and control signals in digital form.

A digital / analogconverter converts the audio signals and amplifies them in the bridge end stages. The output power of the mid-range loudspeaker is 40 W and that of the woofer is 70 W.

Additional features are the central front loudspeaker in the instrument panel, the mid-range speaker in the rear doors and the tweeter under the rear shelf.

The customer can take advantage of a personalised sound adjustment via a 7 band equaliser. Correction is carried out by the controller and the Control Display on the ASK. The amplifiers are controlled by the ASK.

By mounting a central loudspeaker in the instrument a 5-channel space sound, i.e. the so-called Surround Sound, is available for the first time on a vehicle. The amplifiers are always controlled by the ASK.

Audio functions HiFi and LOGIC 7-HiFi

Besides known functions such as volume, fader, balance, treble, bass and GAL (speed-sensitive radio volume control) the two amplifiers HiFi and LOGIC 7-HiFi achieve the following functions:

7-band equaliser (LOGIC 7-HiFi only)

The frequency ranges can be adjusted individually by customers.

Surround Sound (LOGIC 7-HiFi only)

Enabled / disabled by the customer. Surround Sound is obtained in the interior by using at least 5 channels. With an additional loudspeaker in the front between the left and the right channel, a true space sound can be obtained.

The following functions cannot be controlled by the customer, but they are stored in the software of the amplifier.

Level adaptation

Switching between different audio sources (e.g. CC-CD) should produce no change in the volume.
For this purpose, all levels in ASK are processed at a standard level.

Loudness

To improve reception with a low-volume setting, the low frequencies are slightly amplified.

Vehicle-specific equalising

Tuning the acoustics according to the interior of the vehicle.

Speed-dependent equalising (LOGIC 7-HiFi only)

Adjustment of the acoustics to increasing noise emission from the moving vehicle.

Dynamic compression

In a moving vehicle, the usable dynamics is limited in the upper range by the output power of the amplifier and the load characteristics of the loudspeaker. For this reason, a speed-dependent dynamics reduction must be performed.

Frequency-dependent delay (LOGIC 7-HiFi only)

The frequency-dependent delay is used to introduce time corrections on the single speaker units. This ensures that the audio signals are present simultaneously on all speaker units.

Radio system

- General

In the E65 the radio control unit will no longer be a single module. The radio system has distributed functions which are integrated in the corresponding control units, e.g. the display and control function.

The radio is operated via the controller in the central console, the display is in the Control Display located in the instrument panel and the audio signals are controlled in ASK.

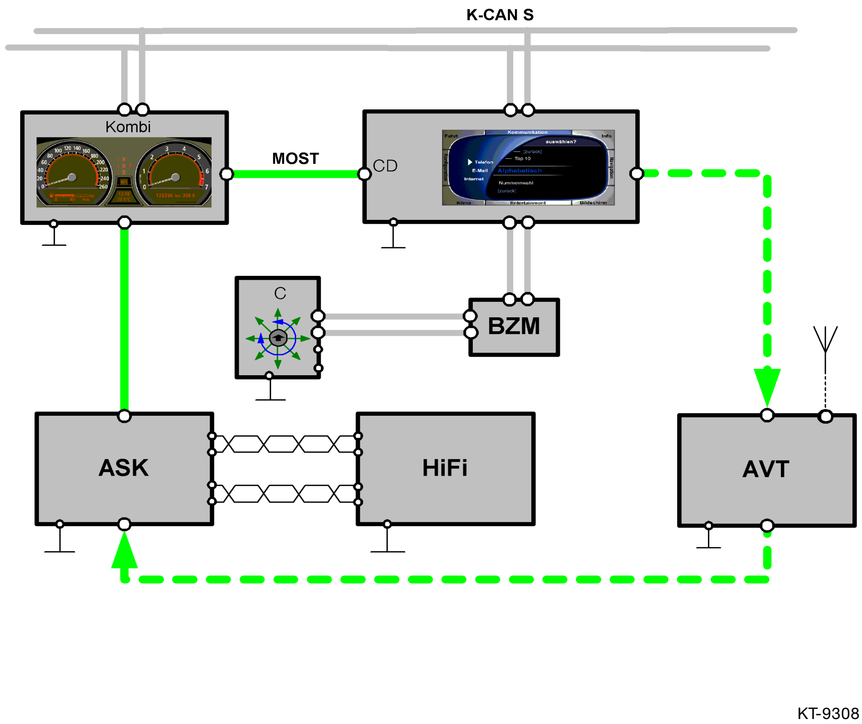


Fig. 19: Radio system with basic equipment

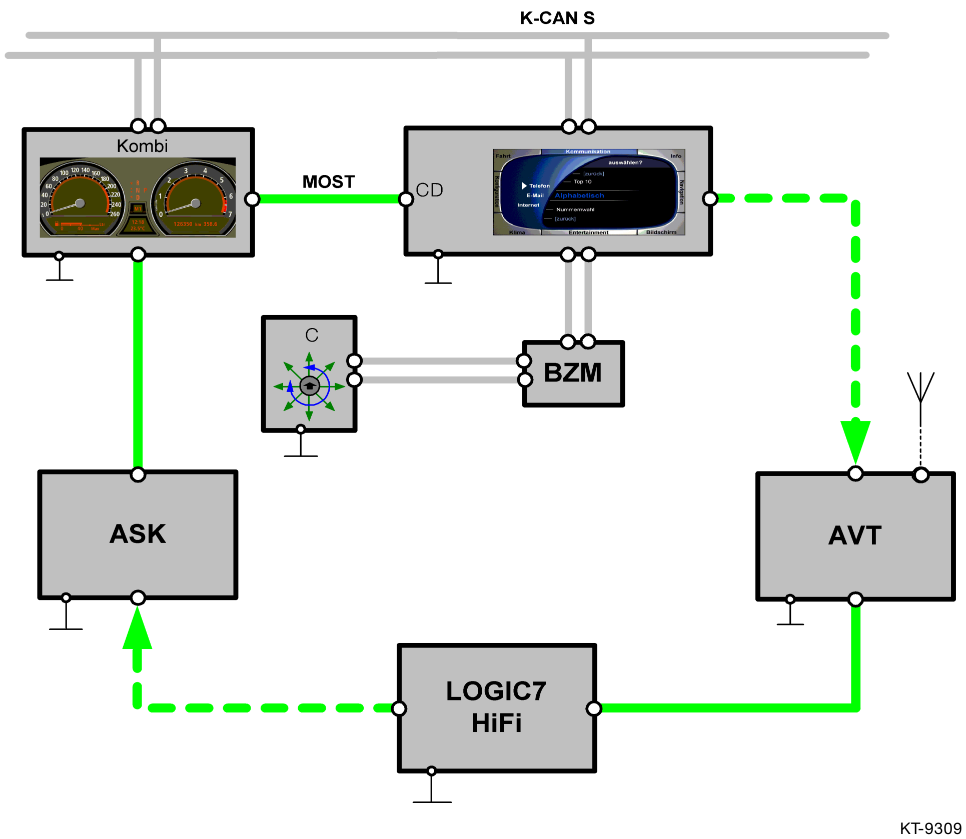


Fig. 20: Radio system with special equipment

Index	Description
▬ ▬ ▬ ▬	The MOST assembly includes other control modules as special equipment.
K-CAN S	Body CAN system
MOST	Media Orientated System Transport
Kombi	Instrument cluster
CD	Control Display
BZM	Control panel module, centre console
C	Controller
AVT	Aerial amplifier/tuner
HiFi	Hifi amplifier
LOGIC 7 HiFi	LOGIC 7 HiFi amplifier
ASK	Audio System Controller

- Tuner

The E65 tuner is different from those of previous car radios. The tuner is actually a control unit with a MOST port. It can therefore be placed in a different location than the radio. The tuner is located in the aerial amplifier.

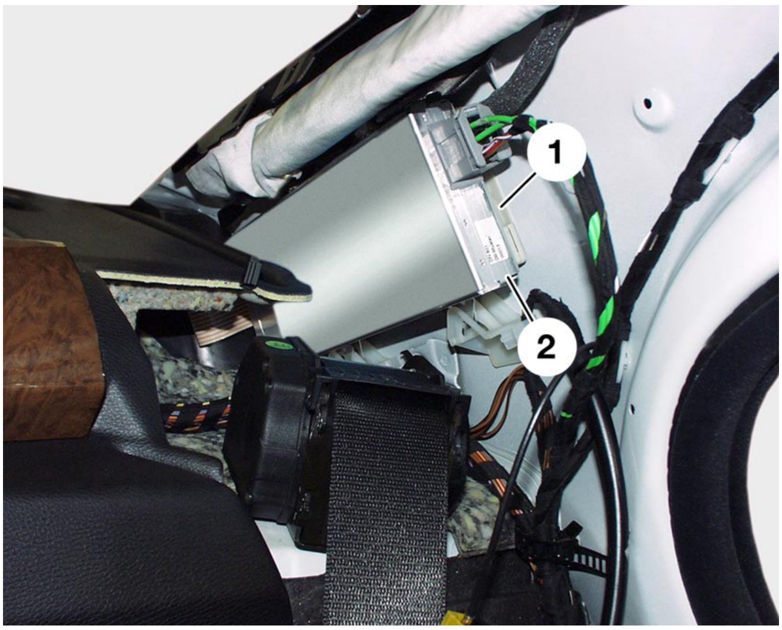
The installation location of the aerial amplifier tuner is on the left rear C-pillar. On the right side, there is an aerial amplifier which is linked via a coaxial cable to the aerial diversity. The tuner provides the power supply to the aerial diversity and the aerial amplifier.

The digital audio signals are transmitted to the ASK via the MOST. Thus, there is no aerial cable. This avoids signal loss and disturbance.

World-wide, there are only two types of tuner.

- Single tuner
- Double tuner

The tuner is coded with the different frequencies, according to country variants.



KT-9217

Fig. 21: Installation location of the aerial tuner: left rear pillar

Index	Description	Index	Description
1	Aerial amplifier with diversity	2	Tuner

Single tuner

The single tuner is a standard tuner which can receive long wave (LW), medium wave (AM), short wave (SW) and frequency modulation (FM) as well as traffic and weather forecast information (US). In the USA weather forecast information is transmitted on 7 different frequencies 24 h a day. The single tuner is available world-wide in all country spec packages as standard equipment except in the ECE version. It is only used in the ECE version if no navigation system is fitted. It corresponds to the former Business radio module.

Double tuner

The double tuner is a standard tuner which can receive long wave (LW), medium wave (AM), short wave (SW) and frequency modulation (FM) as well as TMC data. The double tuner includes a second tuner to receive traffic information announcements on an FM aerial, simultaneously with LW, AM or SW radio. The double tuner is standard in the ECE version in combination with the navigation system or the Professional Audio System.

Frequency ranges

Range	Frequency lower range	Frequency upper range	Application
LW	153 kHz	279 kHz	Long wave radio
MW	522 kHz	1710 kHz	Amplitude modulation (medium wave) radio
SW	5900 kHz	6250 kHz	Short wave radio
VHF	76 MHz	90 MHz	Frequency modulation radio Japan
VHF	87.5 MHz	108 MHz	Frequency modulation radio
VHF	162.4 MHz	162.55 MHz	Weather radio US Version

- Aerial systems

The aerials for radio, television and remote control services (FBD) of the E65 are located in the rear window. They include an AM aerial for LW, MW, and SW and 4 FM aerials for VHF reception. An FM aerial is provided for traffic information for Japan (VICS) and for FBD services at 868 / 433 / 315 MHz.

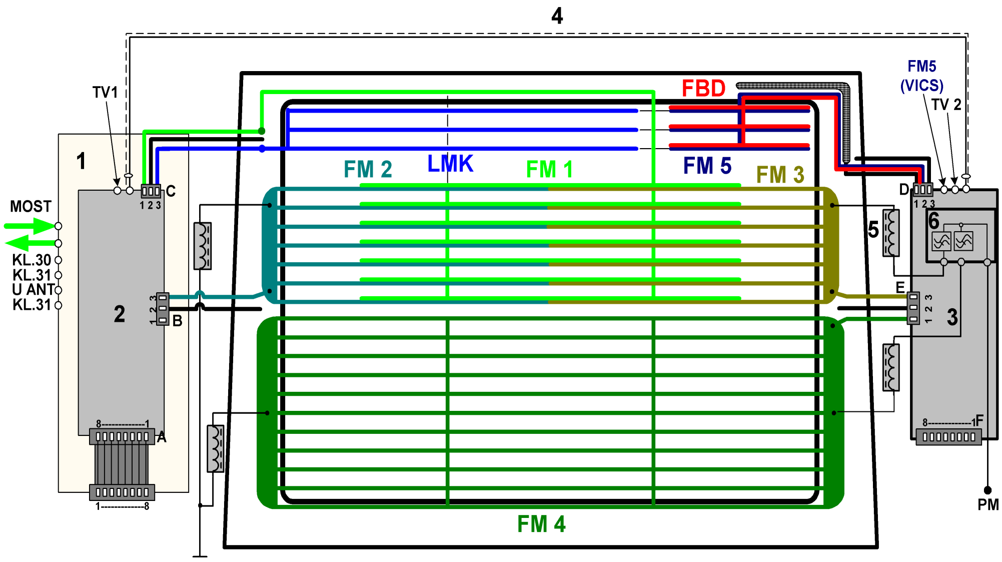


Fig. 22: Aerials for radio reception and wireless services

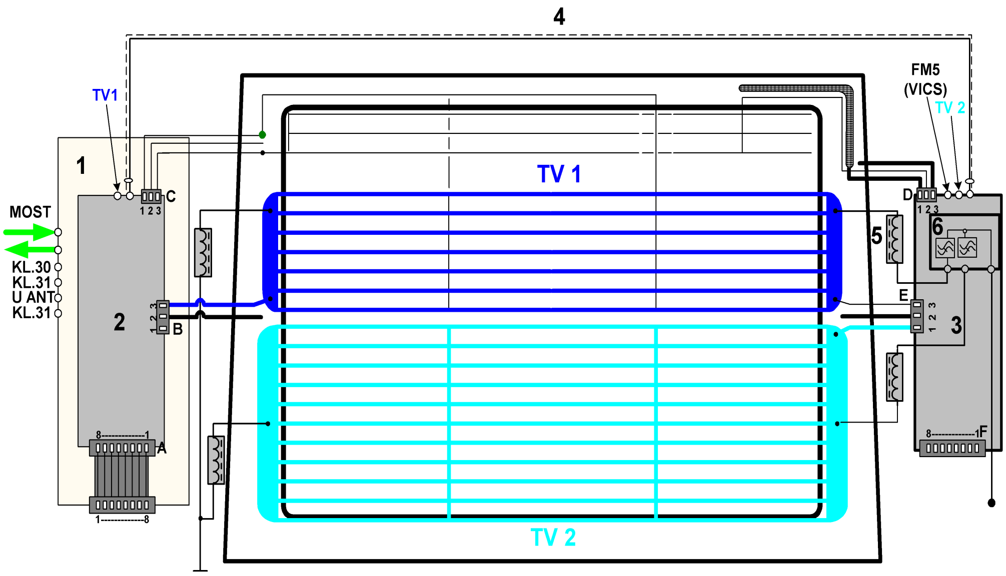


Fig. 23: Aerials for TV reception

Index	Description
1	Aerial tuner
2	Aerial diversity unit
3	Aerial amplifier
4	Coaxial cable
5	Rejector circuit chokes are integrated in the connecting leads
6	Blocking circuit
LMK	AM aerial for long, middle and short waves
FM1	First FM aerial for very high frequency waves
FM2	Second FM aerial for very high frequency waves
FM3	Third FM aerial for very high frequency waves
FM4	Fourth FM aerial for very high frequency waves
FM5	FM aerial for traffic information announcements - Japan VICS
FBD	Aerial for remote control services
TV 1	TV aerial for television reception worldwide
TV 2	TV aerial for television reception worldwide
PM	Power module
KL.30	Power supply, battery positive
KL.31	Earth contact
U ANT	Power supply for aerial modules

Male connector A, ELO 8-pin (aerial diversity)

Index	Description
1	Ground
2	HF signal, AM/FM
3	Ground
4	Power supply for aerial modules
5	DIAGNOSIS
6	CONTROL IN
7	Field strength, LEVEL IN
8	LF signal MPX, AUDIO IN

The pin assignment of the flat cable matches that of the tuner connector.

Male connector B, ELO 3-pin

Index	Description
1	FM 2 / TV 1
2	Ground
3	Not used

Male connector C, ELO 3-pin

Index	Description
1	FM 1
2	Ground
3	AM -LMK

Male connector D, ELO 3-pin

Index	Description
1	Ground
2	FM 5 / FBD 315 / 433 / 868 MHz
3	Ground

Male connector E, ELO 3-pin

Index	Description
1	FM 4 / TV 2
2	Ground
3	FM 3

Male connector F, ELO 8-pin (aerial amplifier)

Index	Description
1	Terminal 30
2	Ground
3	Data OUT
4	RX on
5	Power supply for aerial modules
6	Not used
7	Engine OUT
8	Not used

Audio CD changer

The audio CD changer on the E65 has the same features and mechanical design as its equivalent in the E38 / E39.

This is thus a 6-way-changer with magazine. It is installed on the passenger side in the instrument panel behind a decorative strip of matching colours. If you press the eject key and use the powered flap control, you can control the insertion and extraction from the magazine from the driver or the passenger side.

Fig. 24: Location of the CD changer on the instrument panel support

Index	Description	Index	Description
1	Glove box opening button	3	CD magazine eject button
2	CD changer cover flap	4	CD magazine

KT-9173

Fig. 25: CD magazine push-in compartment

A new feature of the CD changer is the optical connection to the MOST bus. The CD changer is controlled via the MOST bus interface and sends its digital audio signals over it. The audio signals are sent to the ASK which transmits them via the amplifier and the loudspeakers.

To operate the CD changer, you can use the following options:

- Controller and Control Display
- ASK and multifunctional steering wheel
- Voice input

The CD changer offers the following functions:

- Play
- Fast Forwards and Fast Backwards
- Music track search (Skip)
- Scan
- Random
- CD display

The Scan and Random functions only apply to the current CD, not to the whole magazine.

Diagnosis

The CD changer has diagnosis capability. The CD changer offers the following diagnosis functions.

- Start diagnostic mode
- Read out device identification
- Detect and Store the occurring errors
- Output the fault memory contents
- Delete the fault memory contents
- Read out the serial number
- End diagnostic mode

Service information

When you replace the CD changer, you must remove the transit screws and cover the holes with the adhesive film provided.

Telephone system

- General

On the E65 a GSM telephone (Global System for Mobile Communications) is offered as country-specific (LA) or optional extra (SA). It essentially includes the fixed equipment already introduced in model year 2001, i.e. the basic interface telephone II (BIT II) with wireless earpiece (SBDH). A tandem telephone system is available as an additional SA for the rear. The maximum transmitting power of the telephone is 8 W.

In the E65 three different country variants are available:

- ECE version at 900 MHz
- US version at 810 and 1990 MHz
- Japanese version at 800 and 1500 MHz

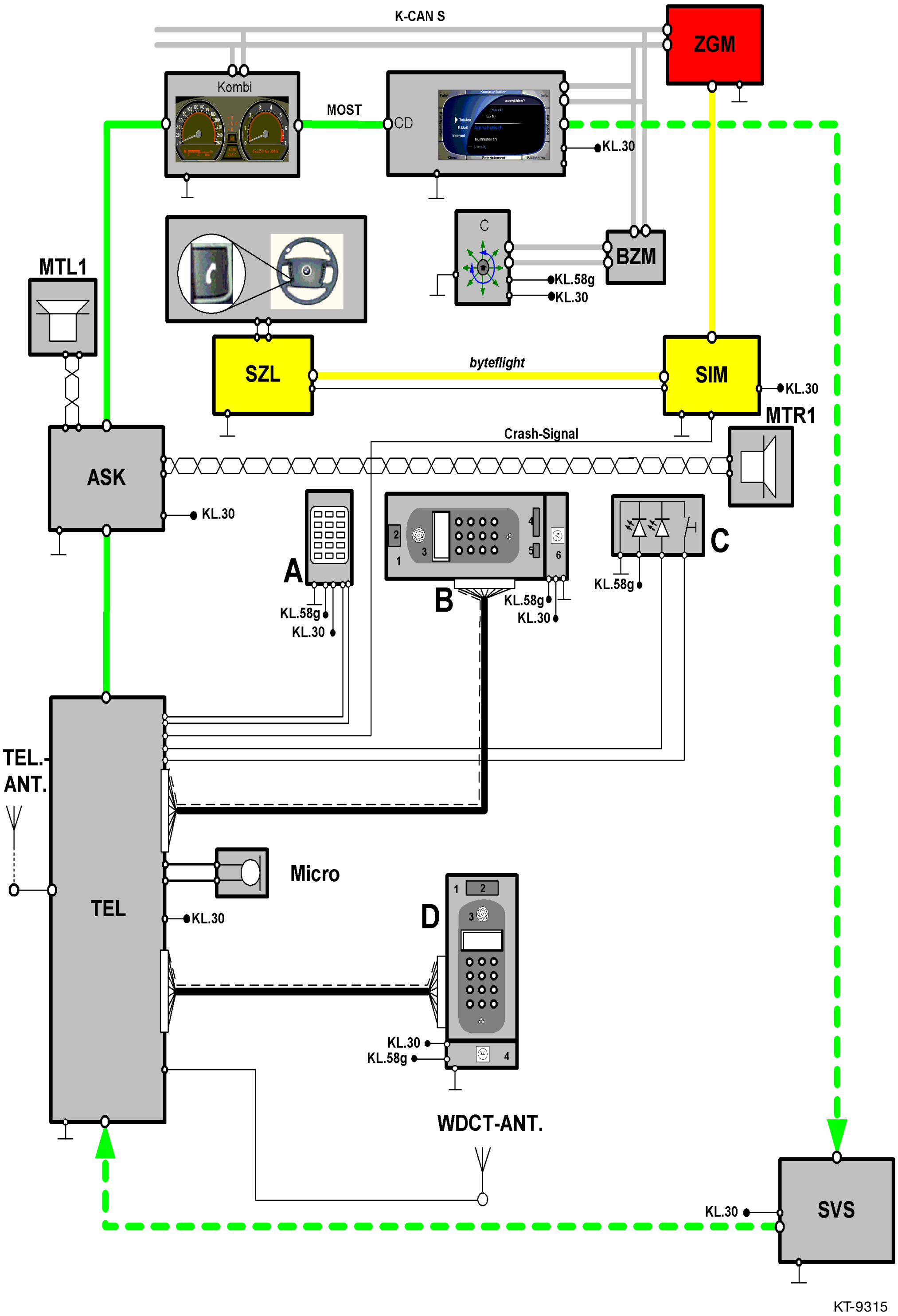


Fig. 26: Block diagram of the E65 telephone system with tandem telephone

Index	Description
	The MOST assembly includes other control modules as special equipment.
Kombi	Instrument cluster
CD	Control Display
ZGM	Central gateway module
SIM	Safety information module
SZL	Switching centre for steering column
BZM	Control panel module, centre console
C	Controller
MTL1	Front left mid-range speaker (output loudspeaker telephone and SVS)
MTL2	Front right mid-range speaker (output loudspeaker telephone and SVS)
ASK	Audio System Controller
TEL	Transceiver unit
Micro	Microphone with hands-free equipment and voice processing system
SVS	Voice processing system
TEL.ANT	Telephone aerial
WDCT	Antenna for cordless handset
A	Phone Board
B	To mount the SBDH
1	Telephone drawer
2	Eject button
3	SBDH (cordless keypad handset)
4	SIM card receptacle
5	Locking button
6	Charging electronics
C	Emergency call button
D	Eject box for rear-compartment telephone
1	Eject box
2	Eject button
3	Rear SBDH (cordless keypad handset)
4	Charging electronics

- Telephone ECE version

In the ECE version you have two options available.
1. The SBDH is located on the front in the telephone drawer.
2. A tandem telephone system is available as SA with a front SBDH and an additional SBDH in the rear central armrest.

To integrate the telephone in the communication assembly, the transceiver is equipped with a MOST port. All data are output in digital form via the MOST assembly, e.g. the voice input for the voice processing system uses the hands-free microphone of the telephone. In the transceiver units the analog signals from the microphone are converted into digital signals and forwarded to the voice processing system via the MOST port.

The telephone can operate with the following systems:

- Cordless keypad handset
- Controller
- Phone Board
- Voice processing system

- Component overview and functional description

The telephone system is made up of the following components:

- Telephone drawer
- Front cordless keypad handset
- Cordless keypad handset and eject box for the rear (SA)
- WDCT aerial
- Transceive unit
- GSM aerial
- Hands-free microphone
- Audio System Controller for hands-free set
- Emergency call button
- Voice processing system (SA)

Telephone drawer

In the E65 the cordless handset SBDH is located in a drawer below the ASK and the phone board. The drawer opens by pressing a pushbutton. The SBDH is located with the display in the lower part of the drawer and is secured to it.

Fig. 27: View of the telephone drawer with SBDH

Index	Description
1	Eject button for the SBDH
2	SBDH (cordless keypad handset)
3	Slot for the SIM card holder
4	Eject button for the SIM card holder
5	Opening button for the telephone drawer

The SBDH is released by an eject button on the left of the drawer. The SBDH is lifted by a mechanism so that it can be withdrawn.

The SBDH is linked via charging contacts to the charging electronics, which is located on the right of the drawer. The SIM card reader (Subscriber Identity Module) for the plug-in SIM card (3 V) is located on the right of the drawer.

You can extract the SIM card holder with the SIM card by pressing a pushbutton.

The connections of the telephone drawer in the vehicle wiring harness are given in the following table:

Pin no.	Description	Remark	Signal	Input / output
1	Terminal 30	Continuous positive	Analog	Input
2	Terminal 31	Ground	Analog	Input
3	Terminal 58g	Unassigned	Analog	Input
4	Tel. ON	System active	Analog	Input
5	Unassigned			
6	12 V	Switch on the power	Analog	Input
7	IN	SIM card inserted	Digital	Output
8	RES	Reset	Digital	Input
9	DATA OUT	Data port to BIT II	Digital	Output
10	DATA IN	Data input from BIT II	Digital	Input

Tandem telephone system

With the tandem telephone system option, the second SBDH is placed in a second eject box in the central arm rest. Both SBDHs are absolutely identical.
The eject box is the same previous BIT II version, installed in the central console.

Fig. 28: Rear-compartment telephone in the central arm rest

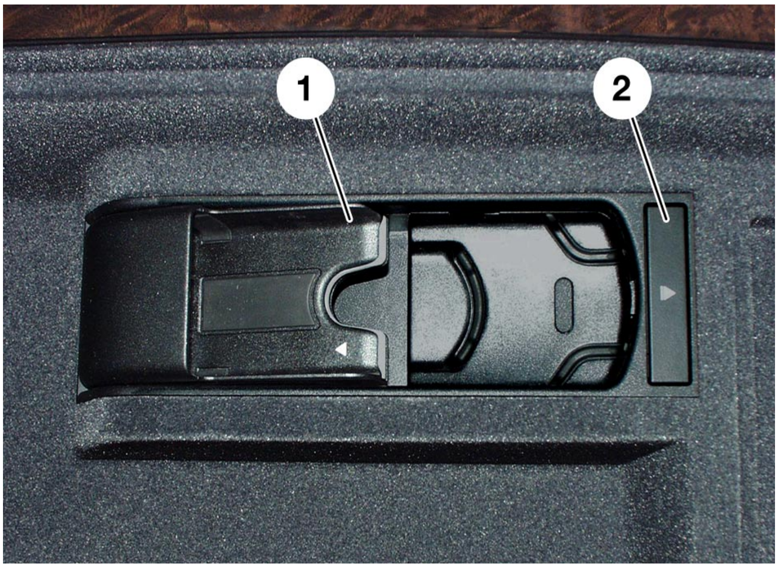


Fig. 29: Eject box of the rear-compartment telephone

Index	Description	Index	Description
1	SIM card reader	2	Eject button

The eject box contains a card reader for the normal SIM card (3 V) with a large SIM card holder. If both SIM card holders (card reader) are used, the following sequence applies.

The rear card (executive) has priority over the front card (chauffeur).

The cordless keypad handset can be placed in the cradle with the keypad facing up or down. A charging electronics is integrated in the eject box for the purpose of charging the rechargeable batteries in the SBDH.

SBDH (cordless keypad handset)

Component description

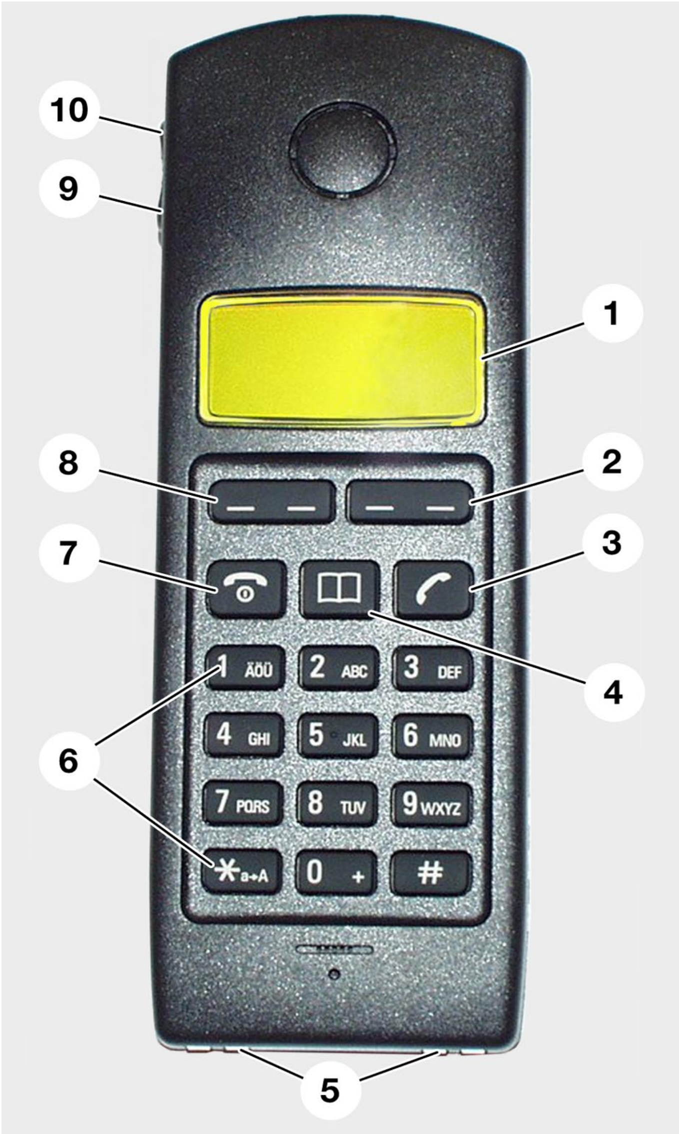


Fig. 30: SBDH (cordless keypad handset)

Index	Description	Index	Description
1	Display 35 x 97 pixels	6	12-key keypad
2	Multifunction key (softkeys)	7	End key
3	Call key	8	Multifunction key (softkeys)
4	Phone book key	9	Receiver volume low
5	Charge contacts	10	Receiver volume high

Features	Data
Range	10 m (always dependent on local conditions and may have a greater range)
Call time	> 3 hours with fully loaded batteries
Standby time	48 hours
Timer (run-down period)	max. 60 min. This time may be adjusted between 0 and 60 minutes
Rapid charge	after 1 hour rapid charging a call of at least 30 minutes is possible
Weight	< 200 g
Rechargeable battery	1300 mAh NiMH 1.2 V Mignon LR6 AA

Functional description

The functions are very similar to the Siemens Gigaset home telephone and the Telekom Sinus model. The transmission frequencies of the home telephone and cordless keypad handset differ.

Radio transmission takes place on the ISM band (Industrial, Science, Medical Band). The BMW system operates at a frequency of 2.45 GHz (DECT standard at home: 1.8 GHz). The ISM band allows transmit powers up to 10 mW. The transmitting and receiving power of the SBDH to the transceiver unit is approx. 10 mW. Due to the low output, the effective range in the vicinity of the vehicle is limited. The aerial under the rear shelf should not be screened by obstacles.

The WDCT (Worldwide Digital Cordless Telephone) protocol is used for voice/data transmission between the base station and SBDH.

Telephoning with the SBDH

For telephoning with the SBDH, the following options are available:

- Start and end the call with the SBDH
- Transfer the call from the SBDH to the hands-free equipment
- Take over the call from the hands-free equipment to the SBDH
- Connect front and rear SBDH units
- Conference between front SBDH, rear SBDH and external parties

If a call is received on the SBDH, press the Send button or the "Accept" softkey to establish the connection. To end the call, press the End button.

If the call was taken on the SBDH, press the "Hands-free" softkey to transfer the call to the hands-free equipment.

A call via the hands-free facility can be accepted by the cordless keypad handset (SBDH) by pressing the Send button and confirming by "YES" or by pressing a second time on the Send button. A call is terminated by pressing the "Terminate" softkey or pressing the End button.

If latched in the telephone drawer, the cordless keypad handset is also activated when the ignition is switched on.

The front SBDH is mounted in the telephone drawer with the keyboard turned downwards. The keypad lighting is not active with the handset in this position or in the "vehicle lighting on" status.

The keyboard lighting is activated when you remove it from the telephone drawer and it remains in this status for 15 s. The keyboard lighting remains on for 15 s after the last keystroke, if sufficient battery capacity is available, otherwise, this duration is shortened.

When the telephone system switches off (terminal R off + delay elapsed) the cordless keypad handset is also switched off by means of a radio signal. The system then goes to sleep mode.

Login and logoff of SBDH

The SBDH is assigned at the factory to the transceiver unit.

If a new SBDH is switched on, it has not yet been assigned to the system, the message "Please login" will appear in the display.

The login procedure for a new handset is conducted with the DISplus or MoDiC. Use the following sequence:

1. Press "OK" on the SBDH
2. Enter the "0000" unit code
3. Finally, send the logon message via the DISplus/MoDiC.

A maximum of two cordless keypad handsets can be assigned to the transceiver unit. After an SBDH has been logged in, the corresponding information is written to a permanent memory (EEPROM) in the transceiver unit.

If a logged-in SBDH fails, the SBDH data is retained in the transceiver unit.

If a new SBDH is now logged in, the data is stored in the second memory location of the transceiver unit. Then all memory locations now contain the telephone data.

Data is stored in both memory locations when two (e.g. tandem system option) operable cordless keypad handsets are logged in.

In the event of the SBDH failing, a new SBDH can be logged in only when at least one memory location is free in the transceiver unit.

Only all logged in cordless keypad handsets can be logged off with the DISplus or MoDiC, i.e. both memory locations are cleared.

If the SBDH to be logged off is still functional (e.g. only scratched), the logoff procedure can be carried out directly on the SBDH (menu -> local settings -> service settings -> logoff).

Logon:
Only with DISplus or ModiC <u>and</u> SBDH

Logoff:
Via DISplus or MoDiC <u>or</u> SBDH

Battery charging function in SBDH

The charging electronics is integrated in the telephone drawer.

If the SBDH is not latched in the telephone drawing, it will not be charged since the SBDH has no contact with the charging electronics. The driver is therefore advised to latch the telephone drawer in the eject box as a safety precaution while driving.

The electronic charger in the telephone drawer ensures that the rechargeable batteries are only charged when capacity has dropped below 1/3. This function and a charging current of approx. 200 mA effectively avoid the memory effect. A 1 hour long call is possible with the capacity of the rechargeable batteries at 1/3. Batteries are nickel-metal Hybrid batteries (NiMH 1.2 V 1300 mAh Mignon LR6 AA).

Note: Do not use ordinary, non-rechargeable batteries. The charging electronics would try to charge them, which would generate an excess of heat and even create a potential explosion hazard.

The electronic circuitry recognises when new rechargeable batteries are inserted in the SBDH. Consequently the SBDH sends a battery formatting request via the ISM link to the transceiver unit.

The battery capacity indicator on the cordless keypad handset now shows 1/3 full irrespective of whether the rechargeable batteries are discharged or fully charged. The indication will not agree with the corresponding battery capacity before battery formatting has been carried out.

Battery formatting can now be started when the handset is latched in the telephone drawer.

Even if the battery cover is only opened and closed, the electronic charging facility will detect a battery change and request battery formatting.

Note:
The closed-circuit current will be increased by approx. 300 mA while a new rechargeable battery is being formatted.

With a new battery, the formatting run is interrupted. Formatting is continued when the MOST bus is re-activated.

The telephone drawer with its charging electronics, is connected via a 10-pin flexible ribbon cable to the vehicle wiring harness.

Phone Board

The Phone Board is located on the left close to ASK. If you press on the phone board, it moves outwards. The phone board is the control section of the telephone. The ergonomic arrangement in the vehicle allows easy operation of the telephone.

KT-9176

Fig. 31: Phone Board

The connection of the phone board to the transceiver unit is made via a sub-bus, which carries a CAN signal.

The following functions are available to operate the phone board. (see Table)

Key	Short pressure	Key	Long pressure
End	Call termination	End	Turn the telephone on and off
Clr	Delete the last entry/number	Clr	Delete the entire line on the Control display
0	Enter the number 0	0	Enter the + sign
*	Enter the * sign	*	Break for DTMF-selection
#	Enter the # sign	#	Paging (Search on the SBDH)

GSM aerial

The GSM aerial (Global System for Mobile Communication) always include a multiple-band aerial for the telephone and a GPS aerial (Global Positioning System) for the navigation system. For design reasons, the aerial is always installed and colour-coded, even if the optional extras have not been ordered.

KT-9218

Fig. 32: E65 GSM aerial

In the US models two telephone aerials are installed for ranges of 810 MHz and 1990 MHz as well as a GPS aerial. The US aerial is recognisable by its three connections.

Functional description

Both aerial are dual-band and consist of two aerials organised in groups for the telephone transceiver as the aerial for GPS reception.

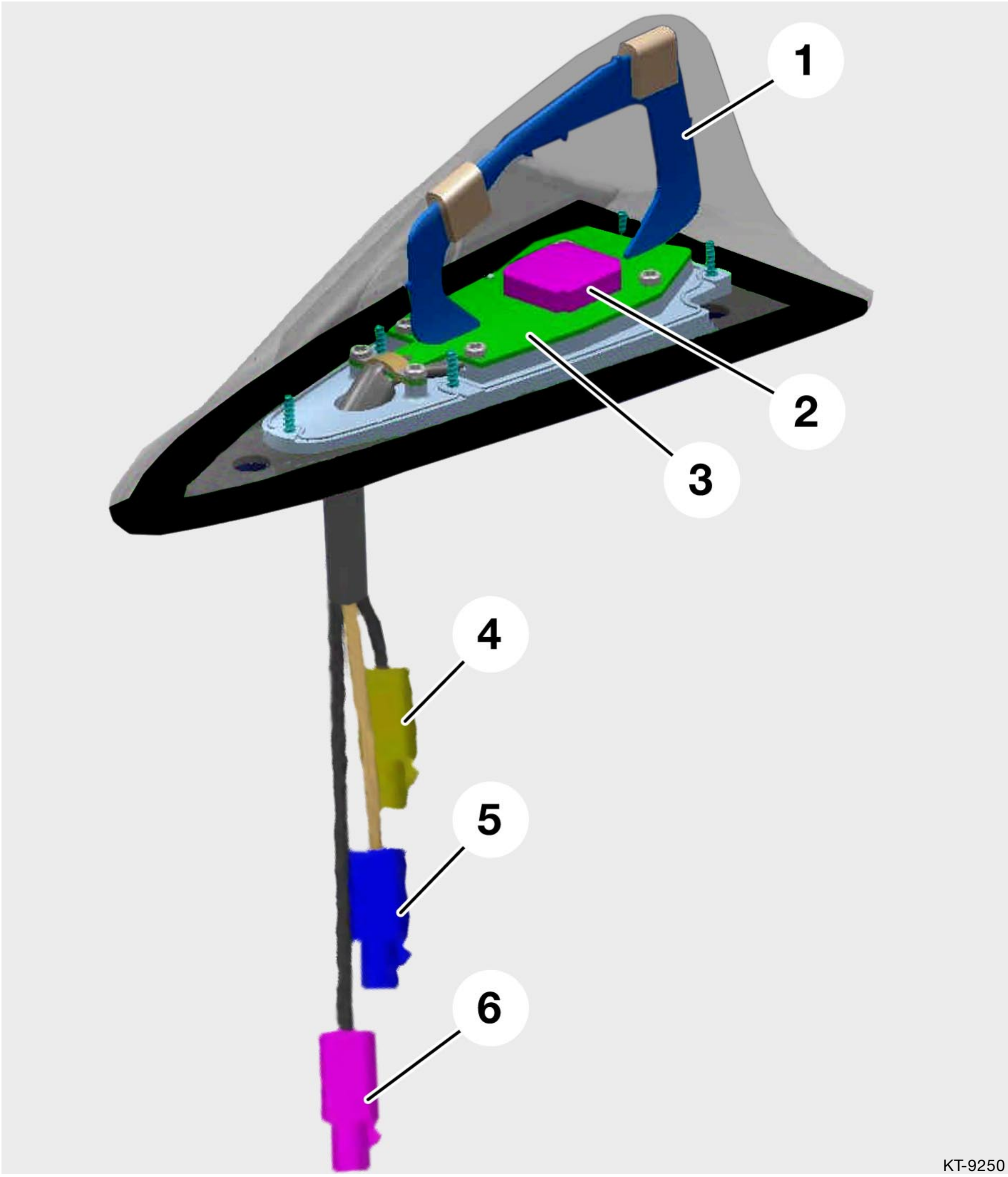


Fig. 33: Inner view of the GSM aerial (US version)

Index	Description	Index	Description
1	Group emitters of the GSM antenna	4	Connection for dual-band (US)
2	GPS aerial	5	Connection for GSM
3	Electronic evaluation unit	6	GPS connector

WDCT aerial

The WDCT aerial (World-wide Digital Cordless Telephone) achieves a connection between the cordless handset SBDH and the transceiver unit.

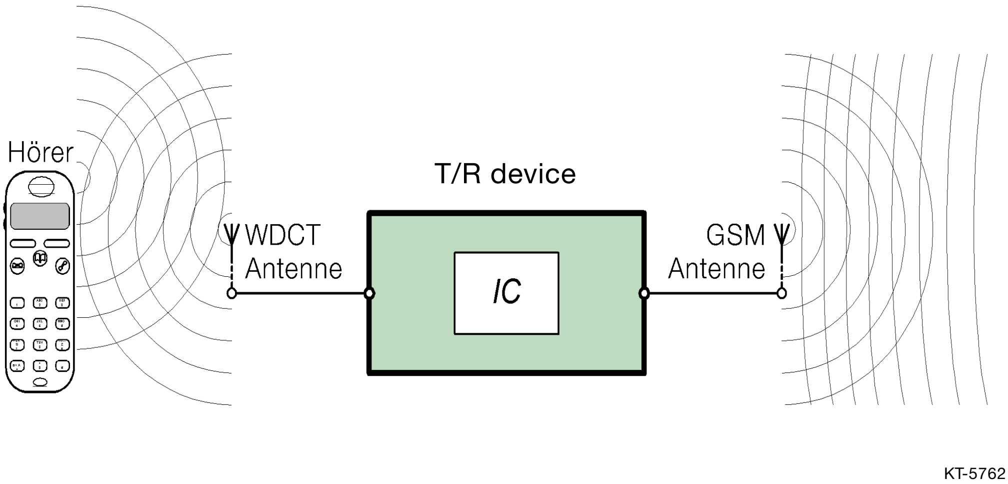


Fig. 34: Schematic diagram, cordless telephone

Index	Description
SBDH	SBDH (cordless keypad handset)
WDCT Antenne	World-wide Digital Cordless Telephone Aerial
T/R device	Transceiver unit
GSM Antenne	Global Systems of Communications Aerial

The WDCT aerial is mounted under the rear shelf. The WDCT aerial is designed for a frequency of 2.45 GHz at 50 Ω. Data is sent and received in WDCT protocol. The operating range of the SBDH is approx. 10 m to the WDCT aerial. Thus you can also phone from outside if you stay in the proximity of the vehicle.

In some country variants, the SBDH cannot be used due to its frequency (2.45 GHz). For this, a WDCT switch function is provided to disconnect the radio interface between SBDH and WDCT aerials. The switch function is included in the vehicle configuration menu of the Control Display.

KT-9168

Fig. 35: WDCT aerial

Note: If the head is turned into the transmission path from the aerial to the SBDH while telephoning with the SBDH, the range of the signal will be shortened considerably. Avoid screening the WDCT aerial with obstacles (cover, screen) on the rear shelf. This would impair the operating range.

Hands-free microphone

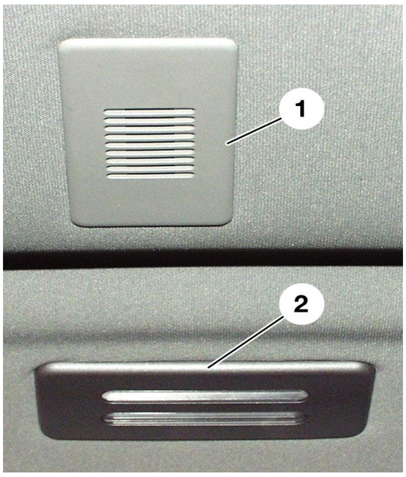


Fig. 36: Hands-free microphone installed in the centre of roof

Index	Description	Index	Description
1	Hands-free microphone	2	Make-up mirror lighting

In the roof liner two covers for the hands-free microphone are mounted outside on the left and on the right close to the sun shades. The hands-free microphone is only mounted on the driver side. The microphone is an active microphone with an input sensitivity of 75 mV/74 dBa.

Functional description

Hands-free operation takes place using the microphone and the installed audio system via the ASK. The ASK provides the audio channels and switches off all other audio sources during hands-free operation.

This is achieved with digital full duplex transmission. It means that the AF (audio frequency) is enabled for both parties. This means it is possible to speak and listen simultaneously. An echo compensation avoids cross-talk attenuation of the calls.

The MOST connection enables the digital signal also for the other MOST equipment connected. The hands-free microphone is used for voice input for the voice processing system.

Differently from the previous system, the signal is not transmitted by the SVS control unit but it remains available as a MOST message on the MOST bus.

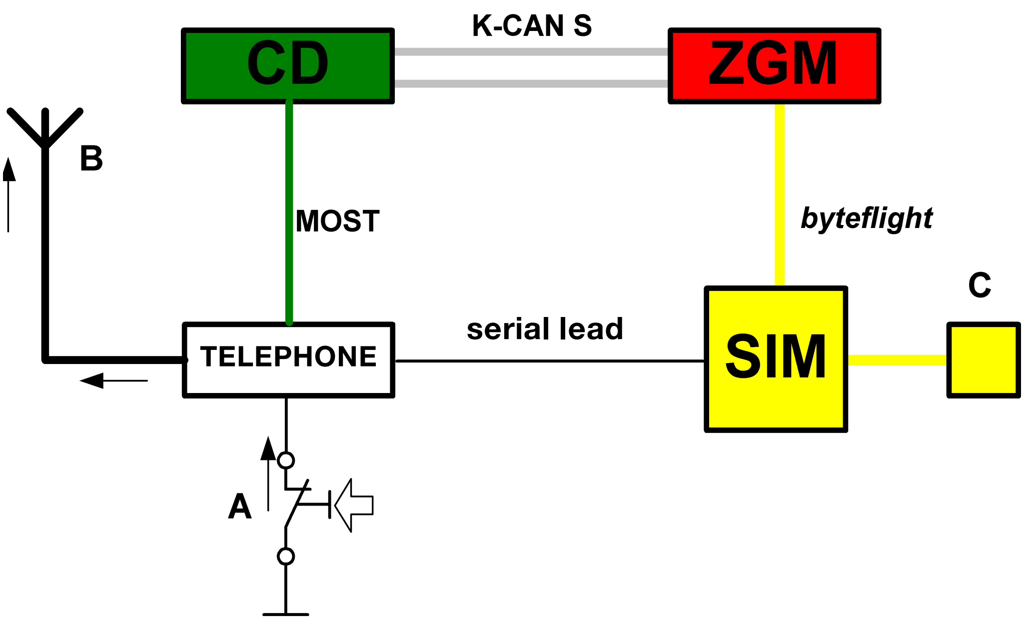
Emergency call button

KT-9180

Fig. 37: Manual emergency call button

Index	Description	Index	Description
1	Emergency call button	2	Emergency call indicator lamp

Manual ermergency call



KT-8530

Fig. 38: An emergency call can be activated manually with the emergency call button.

In an emergency, the driver or passenger presses the emergency call button. This is indicated by a green button LED. The actual position of the vehicle is then displayed on the on-board monitor together with a message to the effect that the emergency call has been transmitted.

The service provider establishes a voice link to the vehicle and asks for confirmation of the emergency call. When the voice link to the provider is established, the green button LED comes on. Following this, emergency action is implemented. This is indicated on the Control Display.

View of control unit

Fig. 39: Aerial connections of the transceiver module

Index	Description	Index	Description
1	WDCT aerial connector	2	Telephone aerial connector

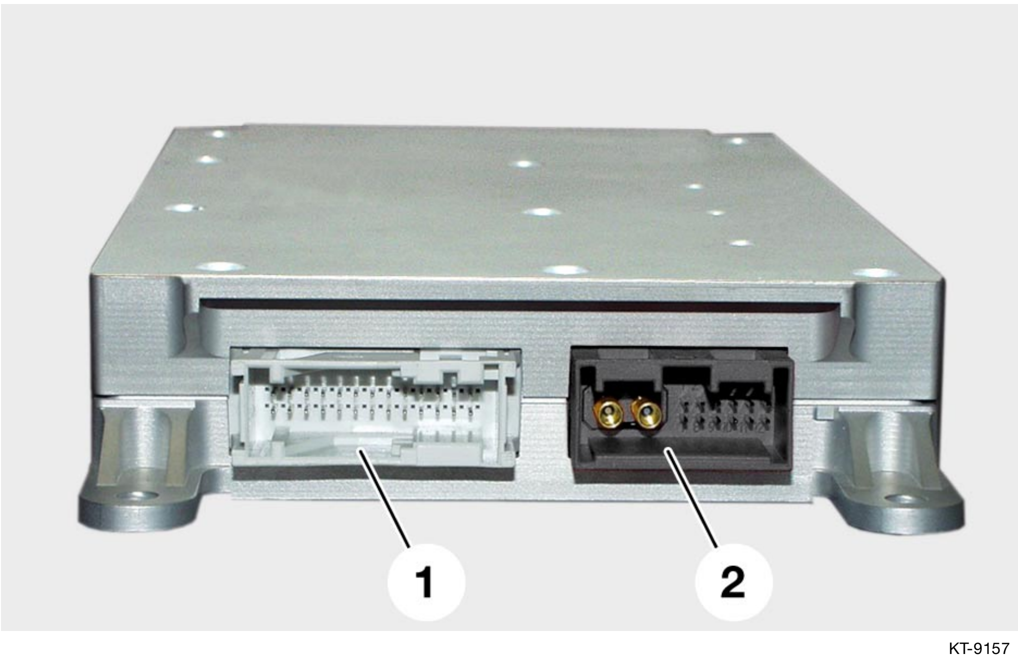


Fig. 40: Connector views of the transceiver module

TV

- General

The TV/navigation functions have been adapted to the new technology on the E65. This model uses the video module 5.

As a result of introducing the MOST bus, it has been necessary to develop a new video module (TV tuner). The video module 5 is designed for use in a multimedia environment.

The opportunity has been taken to incorporate technical innovations. As a result, user friendliness has been improved.

The video module is based on the high-grade capabilities of the modules on the MOST network.

The video signals are initially processed and output using the methods previously applied.

- Functions

1. TV reception
2. TV station list
3. Teletext reception (available 3/02)
4. Video signal conversion
5. Control centre for video signals

New displays/station list

Users will automatically be shown an up-to-date list of stations containing only those channels which can be received at the current location.

For stations with a strong signal which also transmit a station ID, the name of the channel will appear (e.g. BBC1, BBC2, ...) instead of the channel number.

The list of stations can be manually updated by the user. This is done by selecting the function "Autostore."

The list is automatically updated whenever a different audio source (e.g. radio or CD) is selected. In that case, the video module receiver is not required and switches automatically to scan mode in order to update the list of stations.

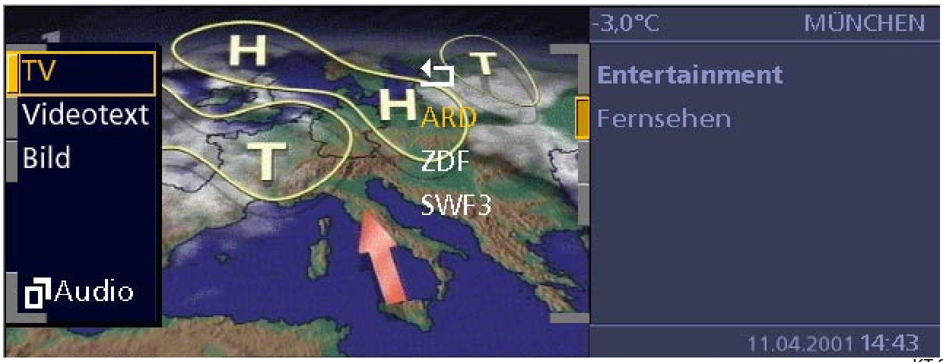


Fig. 41: Display of station list on Control Display

- Variants

There are two variants. They are designated ECE variant and RGB variant. The RGB variant is also referred to as the Japan variant.

ECE variant

The ECE variant is only fitted in combination with an ECE navigation system.

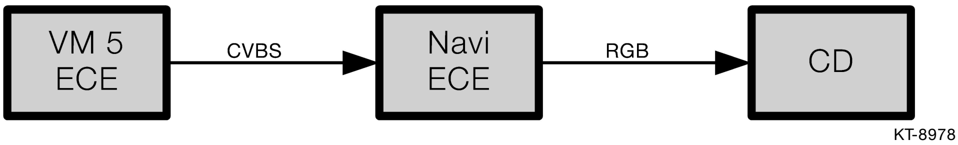


Fig. 42: Schematic diagram of ECE variant

Index	Description
CD	Control Display
CVBS	Composite Video Burst Sync. (colour video, blanking and synchronisation signal)
Navi ECE	Navigation system, ECE variant
RGB	Colour information line for video transmission (red, green, blue)
VM 5 ECE	Video module 5, ECE variant

With this variant, the video signal is sent by the video module using CVBS to the navigation computer. From there it passes via RGB to the Control Display.

RGB variant

The RGB variant is also referred to as the Japanese variant. The RGB variant is always fitted if the vehicle has the Japan navigation system. However, the RGB variant can also be fitted in an ECE vehicle if there is no ECE navigation system fitted.

Example:
If the customer only orders TV (but not navigation system), the RGB variant is fitted.

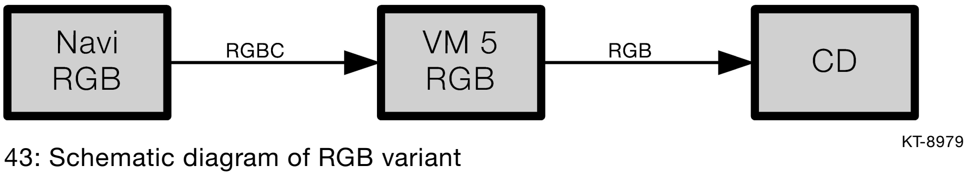


Fig. 43: Schematic diagram of RGB variant

Index	Description
CD	Control Display
Navi RGB	Navigation system, RGB variant
RGB	Colour information line for video transmission (red, green, blue)
RGBC	RGB with an additional lead for a synchronisation signal
VM 5 RGB	Video module 5, RGB variant

With this variant, the video signal is sent by the video module directly via RGB to the Control Display.

- Inputs/Outputs

Inputs

- Two TV aerials
- RGBC (RGB signal lines with an additional lead for synchronisation)
- Terminal 30
- Terminal 31
- MOST
- CVBS (Composite Video Burst Sync.)

TV aerials

The video module 5 has two aerial inputs for TV/teletext reception. The aerials are connected to the video module by co-axial cable. The aerial diversity unit is integrated in the video module.

The aerial diversity unit always switches to the aerial with the better reception. Every 20 ms, the reception quality is checked and the aerial changed over if necessary.

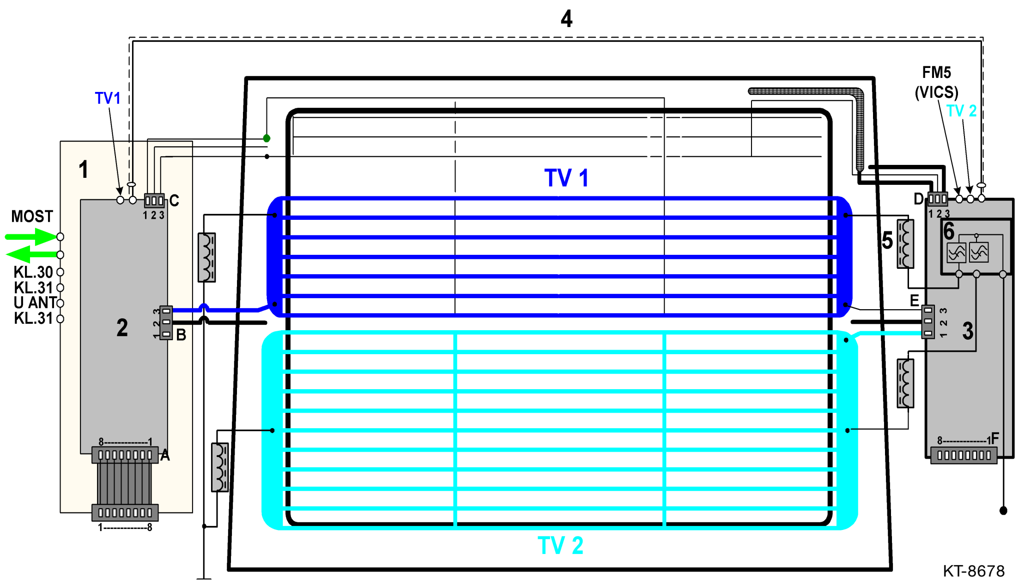


Fig. 44: Aerials for TV reception

Index	Description
1	Aerial tuner
2	Aerial diversity unit
3	Aerial amplifier
4	Coaxial cable
5	Rejector circuit chokes are integrated in the connecting leads
6	Blocking circuit
A,B,C,D,E,F	Connectors; contacts are detailed in the tables below
KL.30	Power supply, battery positive
KL.31	Earth contact
MOST	Media Orientated System Transport
TV 1	TV aerial for world-wide television reception
TV 2	TV aerial for world-wide television reception
U ANT	Power supply for aerial modules
VICS	Traffic information announcements, Japan

Connector A, ELO 8-pin (aerial diversity unit)

Index	Description
1	Ground
2	HF signal, AM/FM
3	Ground
4	Power supply for aerial modules
5	DIAGNOSIS
6	CONTROL IN
7	Field strength, LEVEL IN
8	LF signal MPX, AUDIO IN
The ribbon cable connector pin assignment is mirrored on the tuner	

Connector B, ELO 3-pin

Index	Description
1	FM 2 / TV 1
2	Ground
3	Not used

RGBC

Four leads are used to transmit the 3 colour signals - red, green and blue - and the separate synchronisation signal. The signals are transmitted from the navigation computer (Japan/Korea version) to the video module 5 (RGB variant).

Terminal 30/Terminal 31

The video module is connected to Terminal 30 and Terminal 31 for its power supply. The Terminal 30 power supply comes from the power module.

MOST

The video module 5 is part of the MOST network.

Like all control units in the MOST network, the video module is woken up via the MOST network by an optical signal. The receiver therefore has to be constantly in operation. Nevertheless, the quiescent power consumption is less than 0.02 mA.

CVBS

The CVBS (Composite Video Burst Sync.) lead carries colour and brightness signals as well as synchronisation signals in the case of the ECE variant. The connection is ready for future additions.

CVBS (Composite Video Burst Sync.) = Y/C = S-Video

Outputs

- RGB (red/green/blue colour signals)
- CVBS (Composite Video Burst Sync.)
- MOST

RGB

Three shielded co-axial cables for transmission of the video signal from the video module 5 to the Control Display.

CVBS

The CVBS (Composite Video Burst Sync.) lead carries colour and brightness signals as well as synchronisation signals in the case of the ECE variant. A shielded cable is used. The lead connects the video module to the navigation computer.

CVBS (Composite Video Burst Sync.) = Y/C = S-Video

MOST

The data received along with information for other control units is placed on the MOST bus by the video module 5. Sound, teletext and control signals are transmitted via the MOST bus.

ECE variant

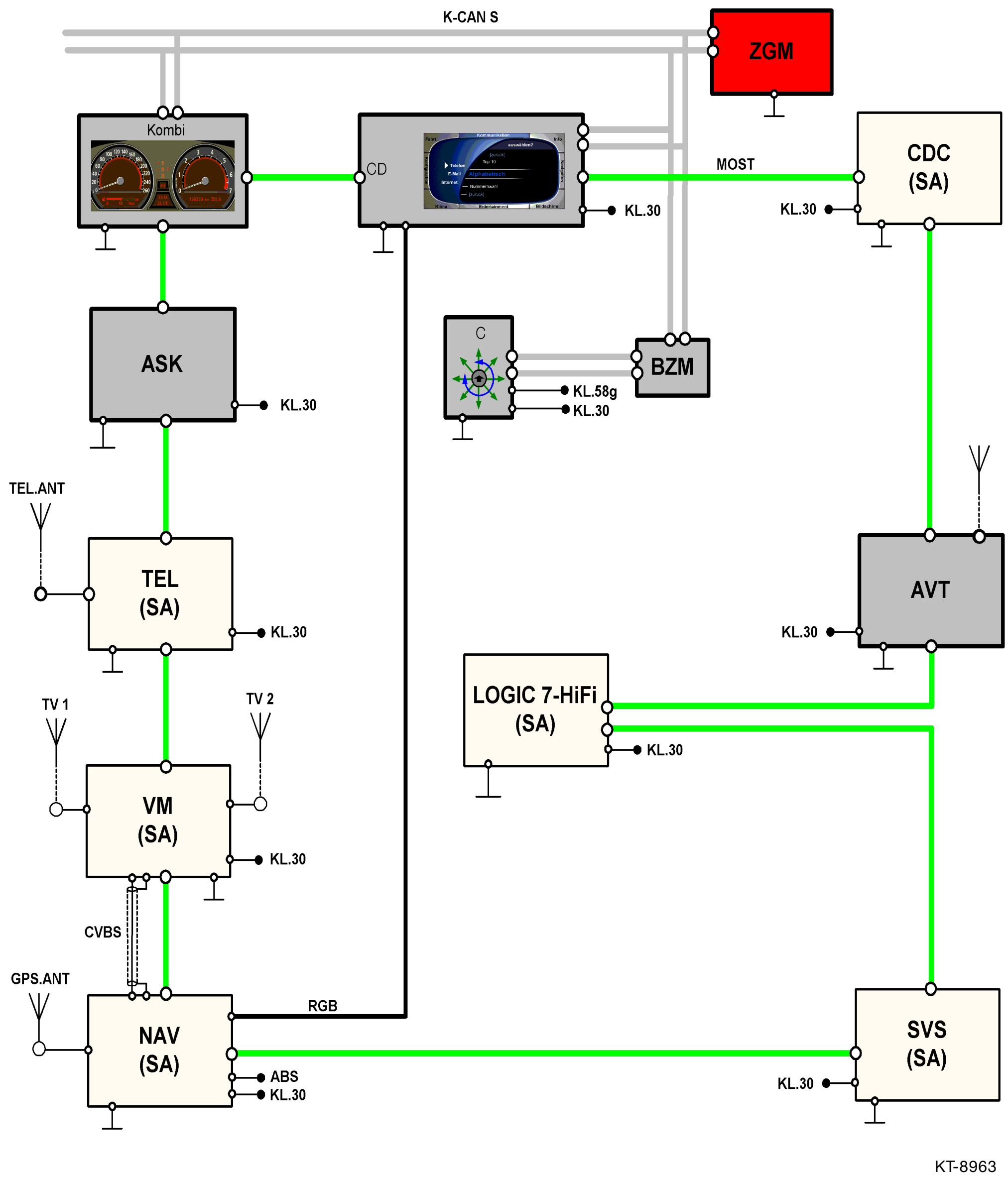


Fig. 45: Block diagram of ECE variant TV/Navigation system

Index	Description
ABS	Wheel speed signal
ASK	Audio System Controller
AVT	Aerial amplifier/tuner
BZM	Control panel module, centre console
C	Controller
CD	Control Display
CDC	Audio system CD changer
CVBS	Composite Video Burst Sync.
GPS.ANT	Global Positioning System Aerial
K-CAN S	Bodyshell CAN system
KL.30	Voltage supply
KL.58g	Lights
Kombi	Instrument cluster
LOGIC 7	TOP HiFi amplifier
MOST	Media Orientated System Transport
NAV	Navigation system
RGB	Colour information line for video transmission (red, green, blue)
SVS	Voice processing system
TEL	BIT II transceiver unit
TEL.ANT	Telephone aerial
TV 1	TV aerial 1
TV 2	TV aerial 2
VM	Video module
ZGM	Central gateway module

- Diagnosis

If faults occur, corresponding fault memory entries can be read off.

Fault memory for MOST

- Receiver failed to pick up a message (Error_NAK)
- Ring break diagnosis carried out (Error_Ring_Diagnose)
- Querying device receives no response although other party is present (Error_Device_No_Answer)

Fault memory for video module 5 diagnosis

- No signal present at RGB input
- No signal present at CVBS input
- Remote supply voltage missing or short-circuited
- One of the aerials not connected or defective
- Both of the aerials not connected or defective
- Tuner 1 defective
- Memory fault
- No signal present at CVBS output 1
- No signal present at CVBS output 2
- No signal present at RGB output 1

Navigation

- Introduction

The navigation computer in the E65 is the platform that carries navigation, telematics and online services.

In terms of the navigation functionality, it more or less corresponds to the Mk-3 navigation computer introduced in model year 2001. A number of additional functions have been integrated, however.

- New, larger screen with an 8:3 size ratio
- Simultaneous map and arrow presentation modes
- Map or arrow presentation in assistance window always possible (simultaneously with radio menu, etc.)
- New-route calculation is faster when the driver has to deviate from the planned route
- Map in anti-aliasing presentation mode (anti-aliasing = method of smoothing edges in graphics)
- Faster on-screen map builds
- "Avoiding toll road" as a new routing criterion
- Multiple destinations can be grouped in a destinations list and picked up one after the other
- Route information arrows, distance and name of the next street shown in the instrument cluster
- Extended voice output ("... follow the M25")
- Message list of traffic interruptions
- Direct destination home (also by voice command)
- Barring of individual roads or entire roadway sections
- Help texts for individual menu items
- Destination by voice input

Like all information and communication systems, the control unit is connected to the MOST bus.

The navigation computer discharges certain controller functions within the MOST network.

- Functions

Navigation display

The information is presented in the Control Display and the instrument cluster. The display window in the Control Display has been resized to a width-to-height ratio of 8:3.

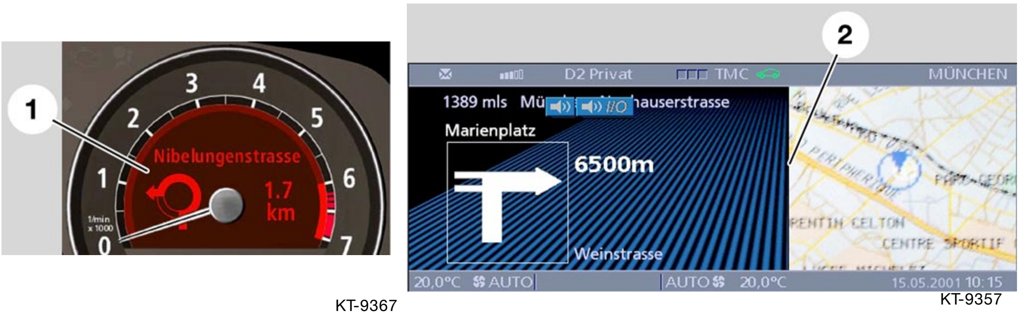


Fig. 46: Navigation display - instrument cluster and Control Display

Index	Description
1	Instrument cluster
2	Control Display

When route guidance is active the navigation display also appears automatically in the instrument cluster. Display of the information in the instrument cluster cannot be deactivated while the route guidance function is active.

The display of navigation information in the Control Display can be deactivated by selecting some other function.

New functions by comparison with the Mk-3 system

- Simultaneous map and arrow presentation modes

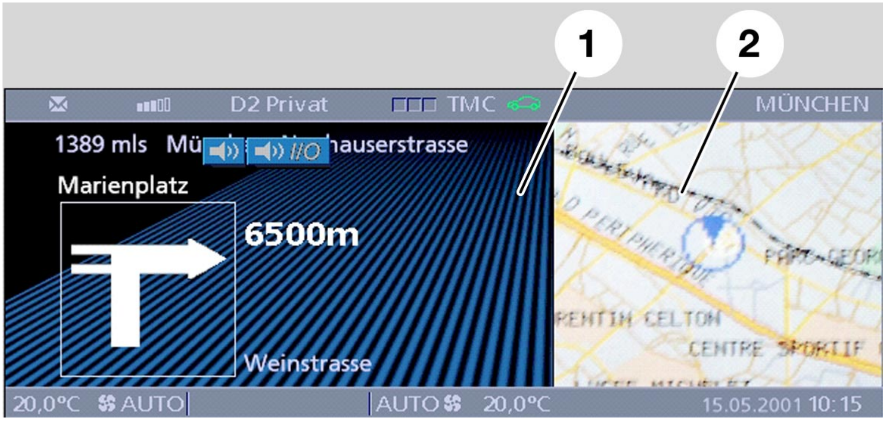


Fig. 47: Simultaneous map and arrow presentation modes KT-9358

Index	Description
1	Arrow presentation mode
2	Map presentation mode

- Map or arrow presentation in assistance window always possible (simultaneously with radio menu, etc.)

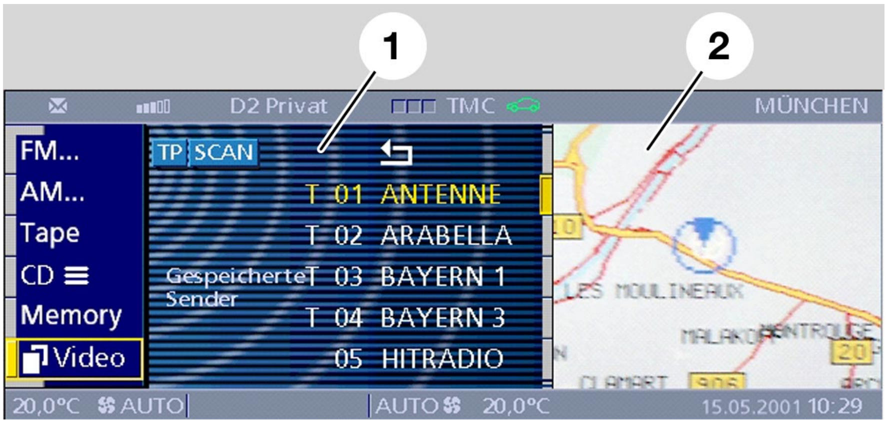


Fig. 48: Map or arrow presentation in the assistance window KT-9359

Index	Description
1	Radio menu
2	Assistance window

- New-route calculation is faster when the driver has to deviate from the planned route

  Instead of recalculating the entire route when the driver deviates from the planned route, the computer merely recalculates only a section of the route before issuing the next instruction, so as to save time.

- Map in anti-aliasing presentation mode

  The map is clearer and the edges are no longer fuzzy.

- Faster on-screen map builds

  Hardware modifications to the navigation computer produce faster screen/map builds.

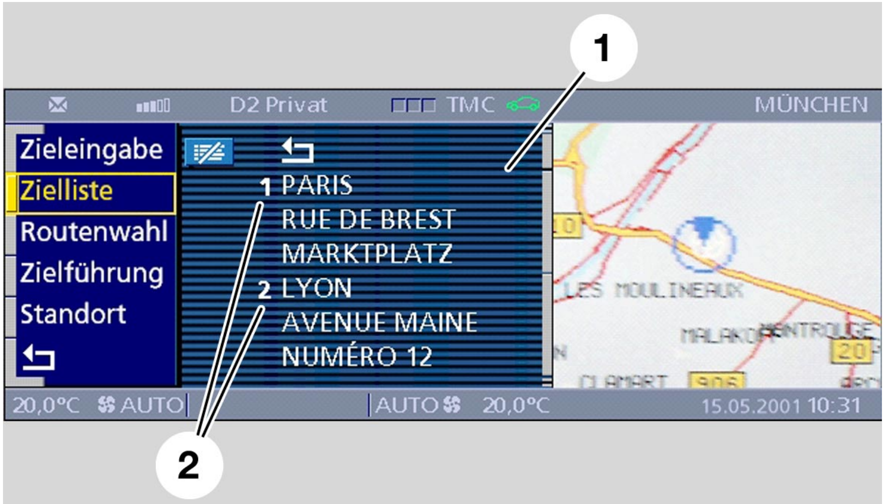
- "Avoiding toll road" as a new routing criterion

Fig. 49: "Avoiding toll road" as a routing criterion

KT-9365

Index	Description
1	Route selection
2	Avoiding toll road

- Multiple destinations can be grouped in a destinations list and picked up one after the other



KT-9360

Fig. 50: Individual destinations/grouped in destinations list

Index	Description
1	Destinations list
2	Individual destinations

- Route information arrows, distance and name of the next street shown in the instrument cluster

KT-9366

Fig. 51: Information displayed in the instrument cluster

Index	Description
1	Name of street
2	Distance
3	Route indicator arrow

- Extended voice output

  For example the following instruction:
  "Continue straight ahead, follow the A9."

- Message list of traffic interruptions

Fig. 52: Message list of traffic interruptions

Index	Description
1	Message list

- Direct destination home (also by voice command)

Fig. 53: Direct destination home

Index	Description
1	Show and change home address
2	Outstanding destinations list is deleted

- Barring of individual roads or entire roadway sections

Fig. 54: Barring - sections/roadways

KT-9363

Index	Description
1	Bar sections
2	Bar route

- Help texts for individual menu items

Fig. 55: Help texts for menu items

KT-9362

Index	Description
1	**Trip** Start route guidance. Destinations in the destinations list are picked up one after the other.
2	Directions

- Destination by voice input

  A destination programmed by voice input can also be called up in the same way.

- Navigation computer/control unit

Building maps on screen is now considerably faster than before. This improvement was achieved by means of modifications to the hardware.

The connection to the MOST bus is by a standardised MOST transceiver. The MOST transceiver incorporates two interfaces (nodes). Each MOST message is subdivided into 3 parts.

- Control data
- Asynchronous data (e.g. navigation system, arrow presentation)
- Synchronous data (e.g. audio, TV, video signals)

A node can analyse only two signals at a time, however, and this is the reason why two nodes are necessary.

In the Control Display self-test both nodes are shown as control units, but only one node can be displayed as a recognised control unit. The second node always has the status "wait" (normally control unit not recognised).

Note:

The Control Display knows the number of MOST nodes, in other words the number of MOST chips in the MOST ring bus. When retrieving the list of MOST control units fitted, the Control Display waits for a response from each MOST node. Normally, each control unit on the MOST bus has one MOST chip. Internally, however, the control unit of the navigation system has two MOST chips.
In response to the query that leads to this list, only one MOST chip answers for the navigation-system control unit. The answering MOST chip is statused as "Navigation," and the other as "wait."
This entry in the list is not an error, but it cannot be suppressed.

- Block diagram

ECE variant

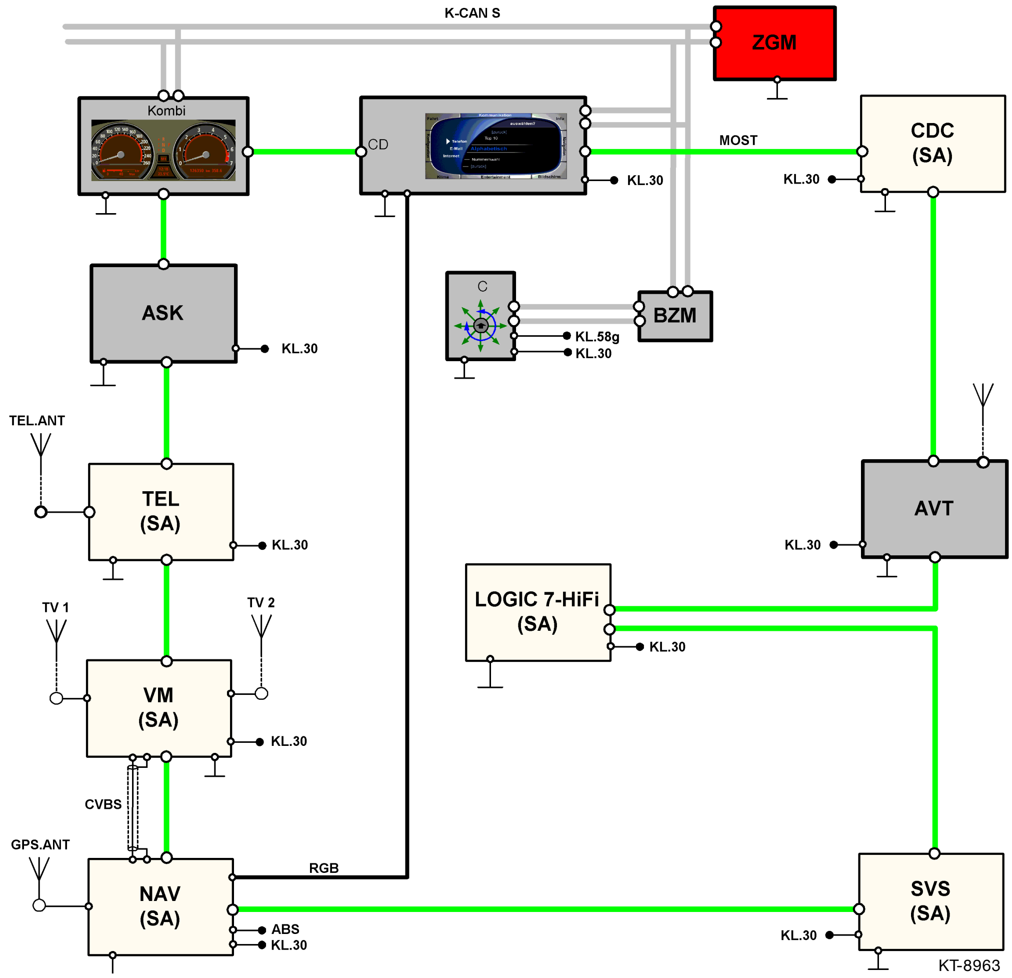


Fig. 56: Block diagram of ECE variant TV/Navigation system

Index	Description
ABS	Wheel speed signal
ASK	Audio System Controller
AVT	Aerial amplifier/tuner
BZM	Control panel module, centre console
C	Controller
CD	Control Display
CDC	Audio system CD changer
CVBS	Composite Video Burst Sync. signal
GPS.ANT	Global Positioning System aerial
K-CAN S	Bodyshell CAN system
KL.30	Voltage supply
KL.58g	Lights
Kombi	Instrument cluster
LOGIC 7	TOP Hifi amplifier
MOST	Media Orientated System Transport
NAV	Navigation system
RGB	Colour information line for video transmission (red, green, blue)
SVS	Voice processing system
TEL	BIT II transceiver unit
TEL.ANT	Telephone aerial
TV 1	TV aerial 1
TV 2	TV aerial 2
VM	Video module
ZGM	Central gateway module

- Inputs/Outputs

Inputs

- GPS aerial
- Speed signal
- CVBS
- MOST

GPS aerial

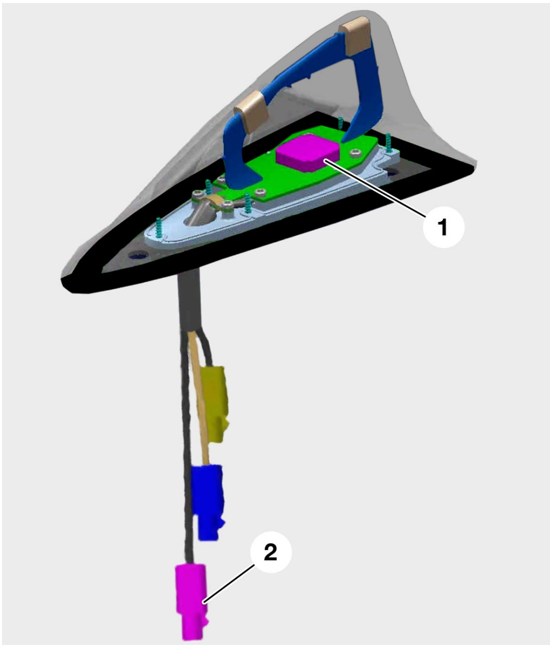


Fig. 57: View inside the GPS aerial

Index	Description
1	GPS aerial
2	GPS connector

Speed signal

The road-speed signal is supplied by the DSC control unit as a processed signal.

CVBS

The CVBS signal comes from the video module. It carries the video signals created in the video module.
The cable is shielded. The navigation computer converts these signals and sends them along the RGB line to the Control Display.
These signals are not processed in the navigation computer.

MOST

The navigation computer is part of the MOST network. Like all the control units in the MOST system, the navigation computer is woken up by light signals on the MOST bus. The receiver therefore has to be constantly in operation. Standby current consumption, however, is less than 0.02 mA.

Outputs

- RGB (red, green, blue signal)
- MOST

RGB

The navigation computer sends the signal for map display directly to the Control Display along the RGB lines.

MOST

The data for the route arrow (vector data) displayed in the instrument cluster and in the Control Display is carried on the MOST bus.

- Diagnosis

If faults occur, corresponding fault-memory entries can be read for MOST only.

Fault memory for MOST

- Receiver failed to pick up a message (Error_NAK)
- Ring break diagnosis carried out (Error_Ring_Diagnose)
- Querying device receives no response although other party is present (Error_Device_No_Answer)

WAP browser

- Introduction

The browser is part of BMW's Online Car project.
It provides the driver with information, services and e-mail.

As this project progresses it will soon provide new and more extensive services, in line with customers' requirements for more information and flexibility in the vehicle.
As of the E65 series production launch, this function will be available as an optional extra and initially in Germany only.
The market launch in all other European countries (with GSM networks) will probably be by the end of 2002.

This application is implemented in the software of the navigation computer (in the same way as telematics, BMW ASSIST).
No hardware components other than the navigation system and the telephone are needed.

The application is started by means of the ASSIST Online function in the BMW ASSIST menu. It establishes online access fro communication between the vehicle and a BMW portal.

A WAP browser is implemented in the vehicle's navigation system control unit. The browser communicates via the MOST bus with the telephone, which acts as a data modem. The MOST interface carries the data to the Control Display (terminal interface) for presentation on the screen.
The browser function can be used when the car is moving or at a standstill.

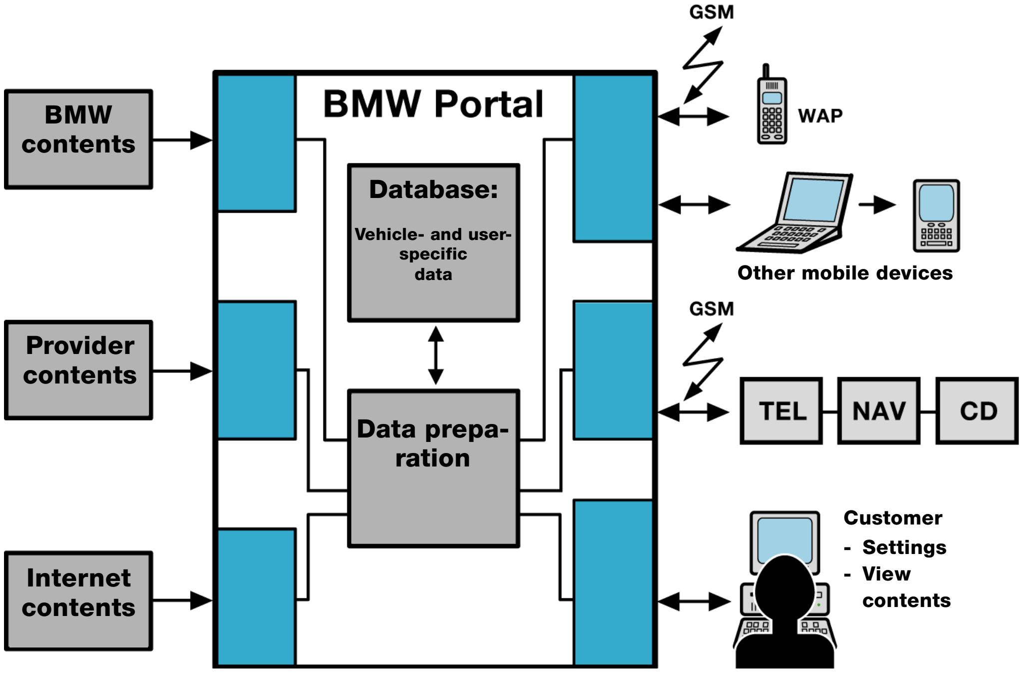


Fig. 58: System overview

Index	Description
BMW Portal	BMW portal
CD	Control Display
GSM	Global System for Mobile Communications
NAV	Navigation system
TEL	Telephone
WAP	Wireless Application Protocol

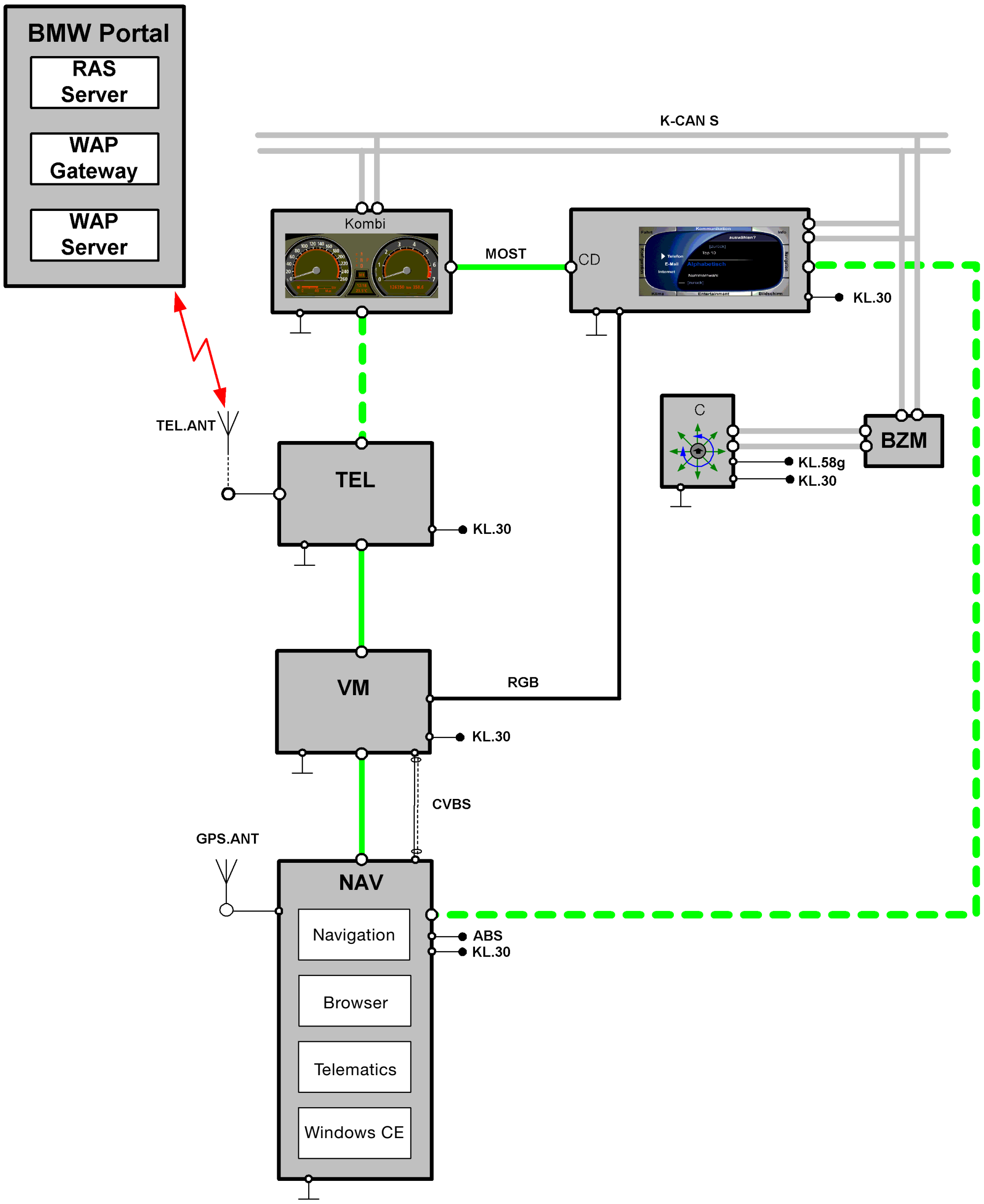


Fig. 59: Block diagram, integration of the browser in "Online Car"

Index	Description
ABS	Wheel speed signal
BMW Portal	BMW portal
BZM	Control panel module, centre console
C	Controller
CD	Control Display
CVBS	Composite Video Burst Sync. signal
GPS.ANT	Global Positioning System aerial
K-CAN S	Body Controller Area Network System
KL.30	Terminal 30, power supply
KL.58g	Terminal 58g, lighting
Kombi	Instrument cluster
MOST	Media Orientated System Transport
NAV	Navigation system
RAS	Remote Access Service
RGB	Colour information cable for video signals (red, green, blue)
TEL	Telephone
TEL.ANT	Telephone aerial
VM	Video module
WAP	Wireless Application Protocol

- General principles of use

On the basis of certain technicalities, the boundary conditions for use of the Online services can be defined as follows:

Layout of the screen

Fig. 60: Layout of the screen

Index	Description
1	Button bar
2	RGB image
3	Assistance window

The screen is divided into 3 windows: button bar (left), RGB image (centre) and assistance window (right).

Assistance window

The assistance window contains information about another application, for example data relating to the navigation system, for example. This window cannot be accessed by the browser, so it is ignored for the purposes of this description.

Button bar

The button bar contains the buttons for navigating with the browser. The following buttons are needed:


KT-9445

Backward: browse backwards through the WAP cards


KT-9446

Reload: reload the current card


KT-9447

Home: load the start page of the portal (currently the login page)

KT-9456

Stop: stop loading or stop the current action

KT-9457

Settings: load a page for changing the browser configuration. "Standard" is the only option available at this time. Its effect is to clear the cache and reset to the factory settings.


KT-9448

Exit: exit the application

It is not possible to browse forwards through WAP cards, so there is no button for this function.

RGB image

The RGB image contains the information obtained by the browser (login information, contents of WAP services, etc.). An RGB object includes links, input fields, buttons for browsing through a card, and softkeys. These elements are known as hotspots.

Softkeys appear at the bottom of the RGB object, not in the button bar. Bear in mind, too, that softkeys are shown only when they are needed.

List of hotspots:


KT-9449

Link


KT-9450

Check-box


KT-9451

Check-box, selected


KT-9452

Radio button


KT-9453

Radio button, selected


KT-9454

Page Up button, selected


KT-9455

Page Down button, not selected

The cursor is represented by a red border.

Cursor and cursor movements

The structure of the cursor on the button bar is defined and cannot be changed.

Turning the controller moves the cursor. The cursor moves in the direction corresponding to the direction of rotation, in other words if you start at the top of the button bar and turn the controller clockwise, the cursor moves to the RGB image and goes to the first hotspot from the top; if you turn the controller counter-clockwise the cursor moves to the next button down.

Only one hotspot is permitted per line, with the exception of the bottom line. This means that the cursor can be moved as if in a circle.

Scrolling

Scrolling is not possible. Its place is taken by page up/down control. If the content of a card is too large to fit onto the screen, it is divided into windows by the browser's rendering engine. Each window can be shown in its entirely on the screen.
The Page Up and Page Down buttons in the bottom line of the RGB image are for switching from window to window. Consequently, the last line of the RGB image cannot be used for displaying information.

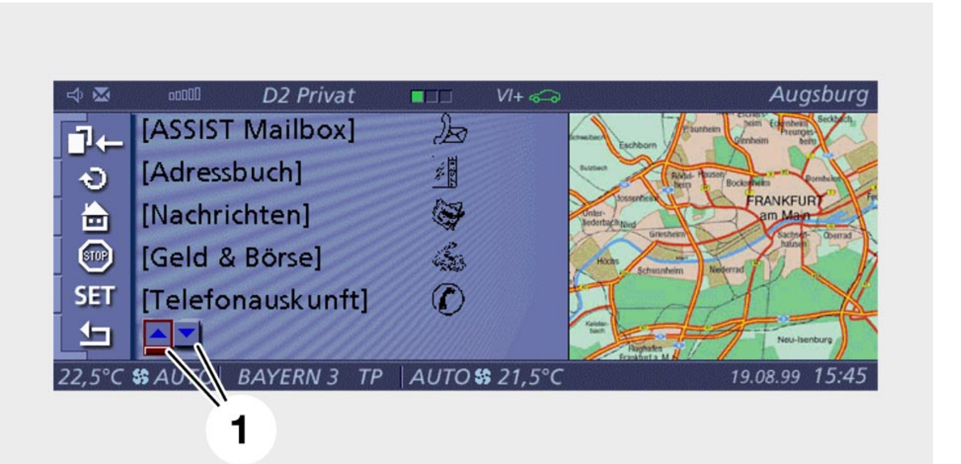


Fig. 61: The Page Up/Down buttons

Index	Description
1	The Page Up/Down buttons

Haptics

The browser tells the Control Display (terminal interface) how many buttons are displayed in the button bar and how many hotspots there are in the RGB image. Only 1 hotspot is permitted per line (with the exception of the bottom line). As many hotspots as fit are permitted in the bottom line.

The controller's haptic depends on the number of hotspots and buttons. The usable medium is the controller. Only rotary and push/press actions are used. The controller can be used as follows:

- **Push:** change to another application
- **Turn:** move the cursor
- **Press:** select a menu item

A push action is used only for changing to another application. Turning the controller moves the cursor from button to button in the button bar and from hotspot to hotspot in the RGB image, and back and forth between button bar and RGB image. Pressing starts the action triggered by a button, actuates a link, or activates an input field.

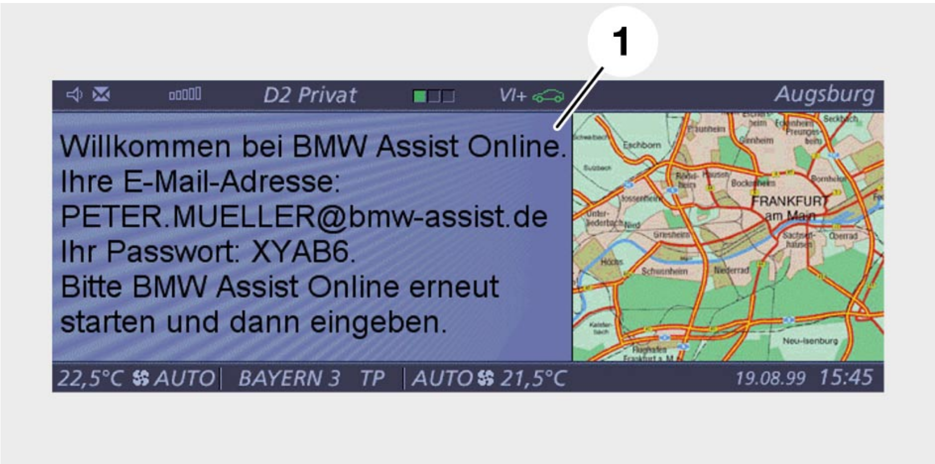
Message boxes

Message boxes are for reporting information and instructions from the application or the portal to the user. Example:

- Unable to connect to the portal
  or
- Password incorrect

The messages can be displayed in two ways:

- In six lines in the RGB object
  or
- To suit the E65 message window as a separate object covering the button bar and the RGB object (see the illustration below). A message window of this nature always has to be acknowledged by pressing the OK button, an action which restores the preceding display.



KT-9422

Fig. 62: E65 message window, button bar and RGB object are covered

Index	Description
1	Message window

Plain-text entries (post code, for example) are needed for certain applications. When entries of this nature are required a virtual typewriter appears on the screen.

Fig. 63: Typewriter and input line

Index	Description
1	Input line
2	Typewriter
3	Please enter the post code

A red square highlights the currently selected character. When it is confirmed the character is shown as a larger font and written into the input line.

Weather information is an example of an application that requires a post code. If the red border on the input field is moved and confirmed, the typewriter opens and the required digits can be selected.

KT-9424

Fig. 64: Weather information

Index	Description
1	**Weather information** Please enter the post code:

- Using the BMW ASSIST Online services

The technical requirements for use of the Online services are as follows:

- Vehicle with ECE navigation
- Permanently installed ECE telephone

The customer must also apply to use the Online services (in the same way as for BMW ASSIST).

The procedure for acquiring approval to use the services is as follows:

1. Customer submits the application
2. The application gives BMW the customer's first name and surname, the VIN, the MSISDN phone number and the list of services the customer wants to use.
3. BMW sends the data to the provider who processes the implementation for BMW. The customer sees only BMW as the service provider.
4. The application is completed.

Service users

Service users can be subdivided into three different categories

- customers
- associate users
- guests

User concept for customers and associate users

When the application is submitted and processed, the customer's data is stored in the database of the portal.

Associated users can only be registered by the customer. The customer can create several associated users and define a custom services configuration for each. Each of these associated users is linked to the customer. The associated users are administrated by means of the PC access to the portal. In addition to access to the portal from the vehicle, the customer also has the options of accessing the portal through a WAP-compatible mobile phone or a PC and calling up the corresponding services.

The associated user must have a different phone number.

User concept for guest users

A guest access is needed for vehicles that are used only occasionally or on single occasions by different persons, as is the case, for example, with rental cars.

In order to log on, however, the guest must be registered with the portal for another vehicle. After logging on, the customer has access to the same services as are available in his/her own car.

Each user has

- a private user name
- a private password
- a private e-mail address
- a telephone number (MSISDN) as per the application

Initial configuration, customer

Initial configuration can be undertaken in the vehicle or with an Internet PC.

In the vehicle

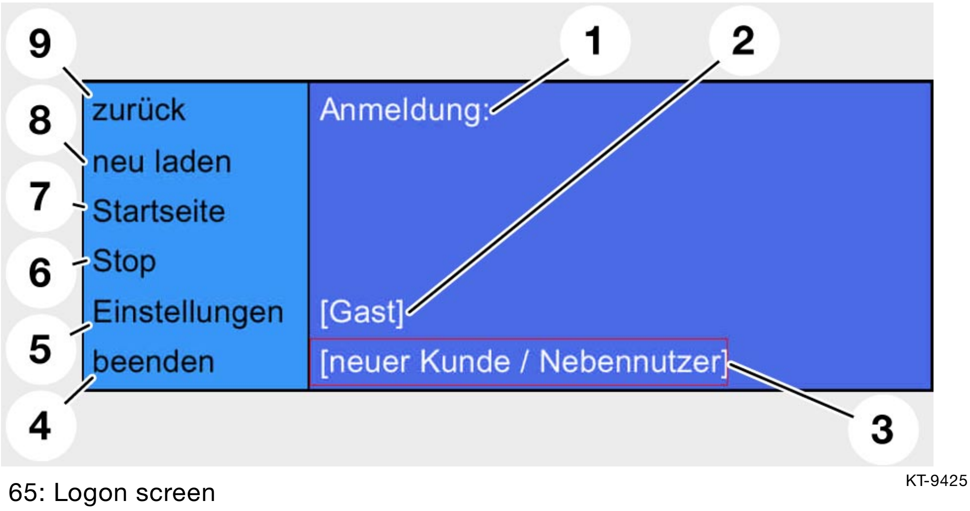


Fig. 65: Logon screen

Index	Description
1	Logon:
2	[Guest]
3	[new customer / associated user]
4	exit
5	Settings
6	Stop
7	Start page
8	reload
9	back

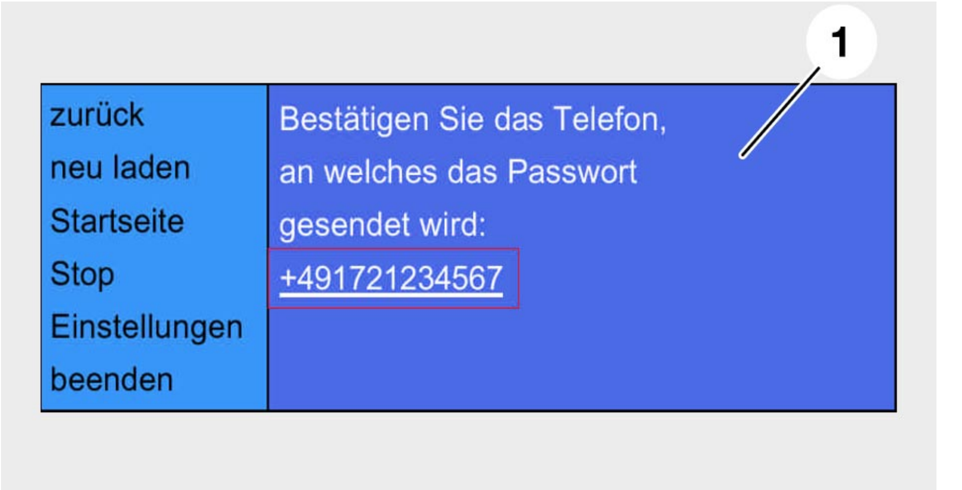


Fig. 66: Confirming the telephone number

Index	Description
1	Confirm the number of the telephone to which the password will be sent: +491721...

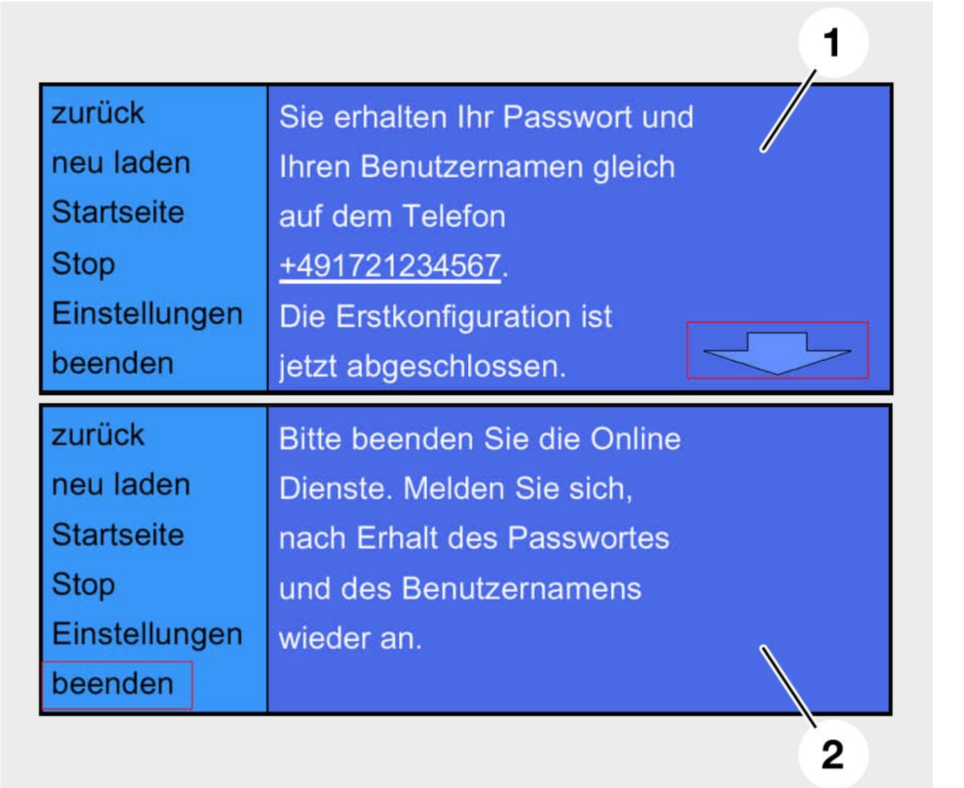


Fig. 67: Message screen

Index	Description
1	Your password and your user name will now be sent to this telephone number +491721... This completes initial configuration.
2	Please terminate the Online services. Log on again when you have received the password and the user name.

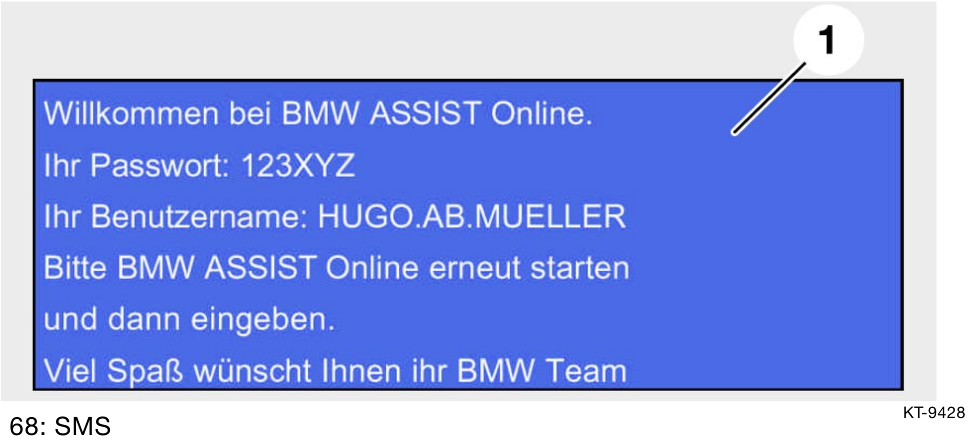


Fig. 68: SMS

Index	Description
1	Welcome to BMW ASSIST Online. Your password: 123XYZ Your user name: HUGO.AB.MUELLER Please restart BMW ASSIST Online and enter password and user name. Best wishes from the BMW team

The dealer hands over the car to the customer. Except for an explicit VIN (Vehicle Identification Number), the vehicle currently has no user specific configuration. The VIN of the associated name and the telephone number of the customer are stored in the portal (after application).

The customer selects BMW ASSIST Online services. The customer's user name is transmitted to the vehicle. If no other users have been defined as yet, only the customer's user name appears in the user list. - "The logon screen is displayed"

The customer selects his/her user name and confirms it.

A cellphone connection is set up (via GSM) to the BMW portal and the VIN is transmitted. The portal searches the database for the customer entry and transmits the customer's telephone number, as stated by the customer in the application, to the vehicle.

Confirming the telephone number
The customer can now select the telephone number and in so doing, confirm initial logon. Further instructions are now issued to the customer.

This procedure ensures that only the registered customer is sent the password. The customer shuts down the browser and, after a short delay, receives an SMS on the phone whose number was specified.

SMS
The customer now has the password and the user name he/she needs to log on to BMW ASSIST Online again and use the online services.

Possible errors:

- Transmission of the password

  - Initial logon is initiated by a person other than the customer. In this case the password is sent to the customer, who cannot use it at this point in time. The portal sets a timer (approx. 3-5 minutes) and the password has to be entered before this timer times out. Logon is cancelled if the password is not entered within this time. The portal cancels the initial logon procedure, so that the true customer has another opportunity to start the initial log on procedure.

  - After the password has been sent to the customer, the password has to be entered within a certain time window (approx. 3-5 minutes). If this is not done, the logon procedure will be cancelled.

- Customer aborts the initial logon process (e.g. by calling up the start page while the logon procedure is still in progress)

  - Portal resets the initial logon
  - Information is sent to user

With Internet PC

The procedure for initial logon through the Internet is as follows:

1. Customer applies for a password and user name

The customer enters the name and network code of his/her telephone card. The customer then enters the phone number of the mobile phone to which he/she wants the password sent. The registration form is sent off when these settings are confirmed.

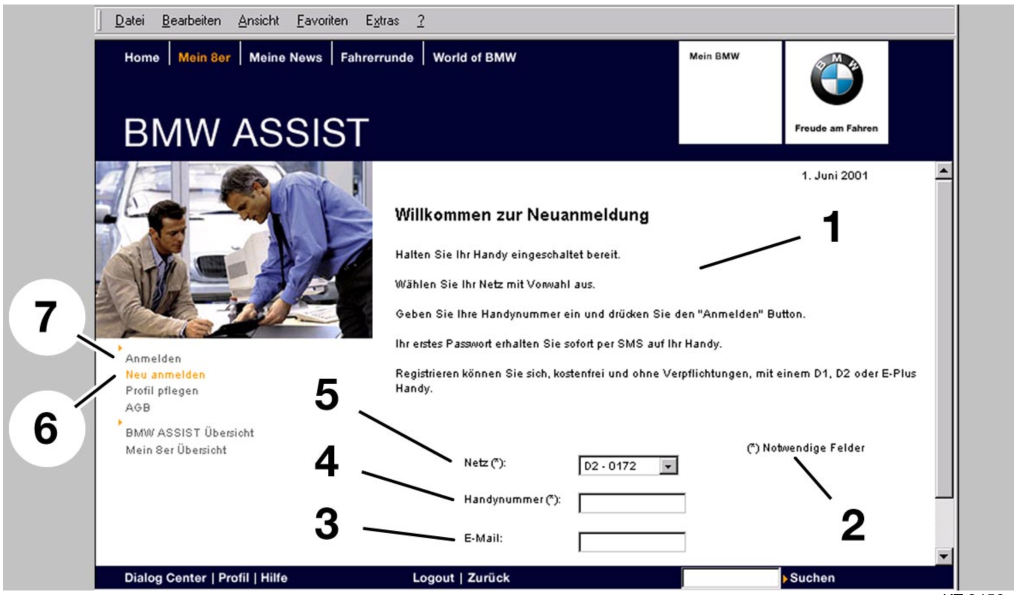


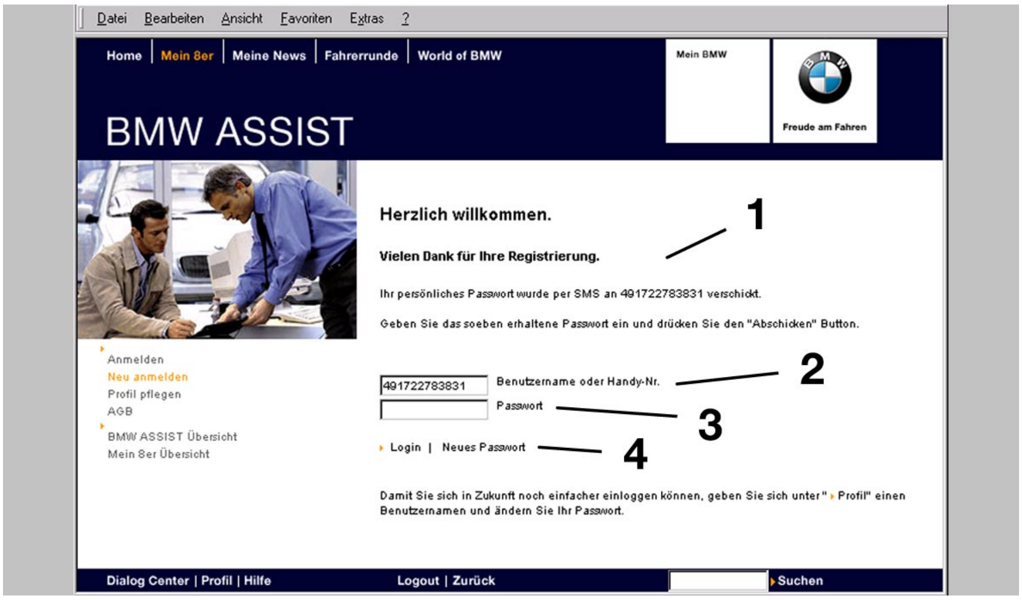
Fig. 69: Registration page in the Internet

Index	Description
1	**Welcome to our registration form** Make sure that your mobile phone is switched on. Select your network and network dialing code. Type in your mobile phone number and click on the "Register" button. Your initial password will be sent right away in an SMS that you will receive on your mobile phone. You can register at no charge and without obligation with a D1, D2 or E-Plus mobile phone.
2	(*) Mandatory fields
3	E-mail
4	Mobile number (*)
5	Network (*)
6	New registration
7	Registration

2. Customer receives password and user name

A password and user name are generated automatically when the portal receives the data (including MSISDN). This information is then sent as an SMS text message to the mobile phone specified by the user. The portal also generates the first start page. This page provides the user with confirmation of successful registration (or indicates that registration was unsuccessful) and enables initial login.

3. Initial login



KT-9460

Fig. 70: Page for initial login

Index	Description
1	**Welcome.** **Thank you for registering.** Your private password has been sent by SMS to 49172.... Please enter the password you just received and click the "Submit" button.
2	User name or mobile phone No.
3	Password
4	Logon I New password

After receiving the password and the user name, the customer must enter the password.

The portal has already entered the user name, so this information does not have to be entered by the customer.
The customer then has to click on the "Submit" button. This sends the login data to the portal.
An e-mail account is created if the data is correct.

The e-mail account contains the first name and family name of the customer. The e-mail account has to have a unique name, so additional letters are inserted.
Example: hugo.ab.mueller@bmw-assist.de.

Once the e-mail account has been created, the portal generates a private start page. The logon process is completed.

Associate users

The customer can define other users, known as associate users, with access to the BMW ASSIST Online services in his/her vehicle. To do so, the customer uses an Internet PC to access his/her personal homepage on the BMW portal and creates a list of registered associate users. The list contains a phone number for each user name. The services can be configured separately for each associate user.

Only the customer can configure associate users; the associate users themselves cannot make changes to their configurations.

When associate users are created the logon list in the vehicle is updated. The procedure for initial logon of an associate user is the same as that for the customer.

Possible errors:

The possible errors are the same as those described for initial logon of the customer.

- Logon

1st possibility

If the customer selects the ASSIST Online services in the display, the screen shown here appears (only if customer is registered):

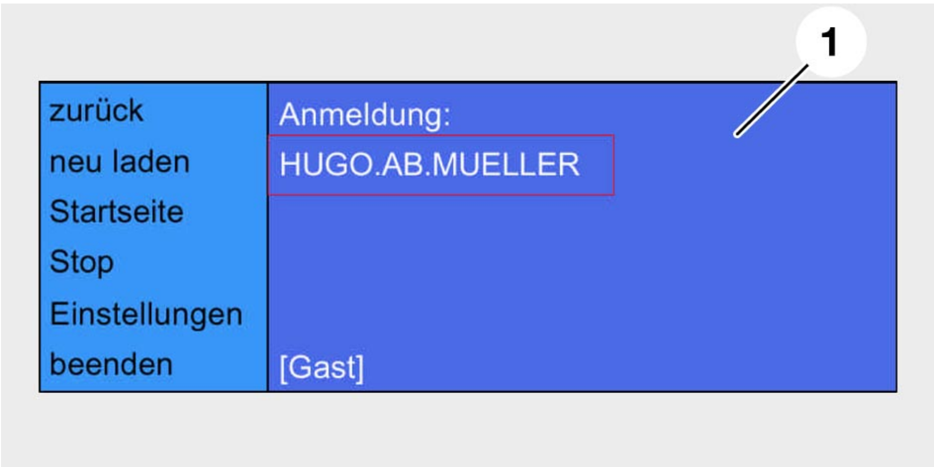


Fig. 71: Logon

Index	Description
1	Logon: HUGO.AB.MUELLER [Guest]

After selecting a name, the customer is prompted to enter the password.

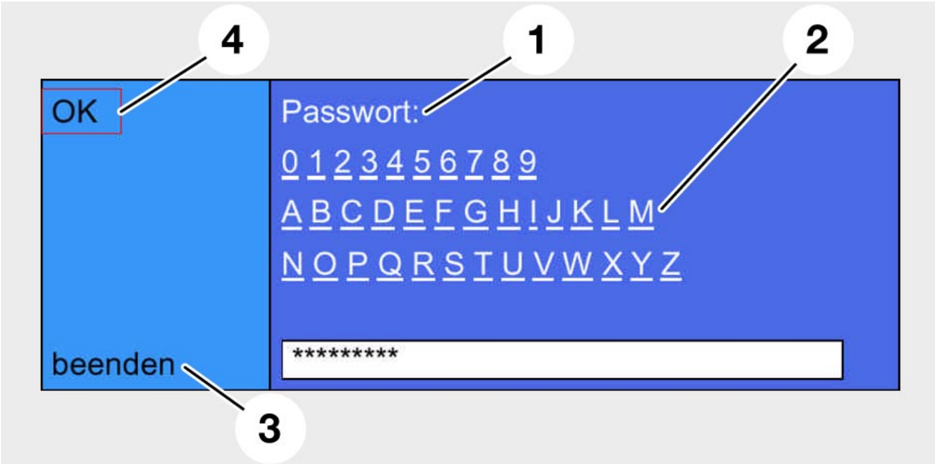


Fig. 72: Typewriter/password

Index	Description
1	Password:
2	Typewriter
3	exit
4	OK

Use the typewriter to enter the password

Click OK to confirm: the password is checked in the portal.

a) Password is correct

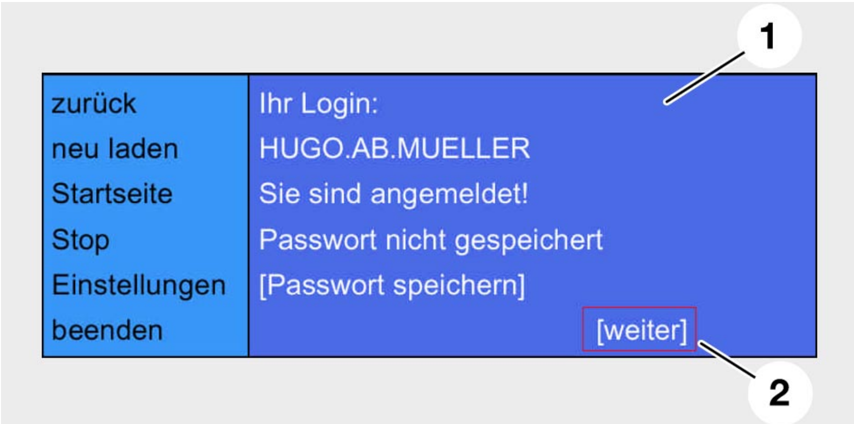


Fig. 73: Confirmation of logon

Index	Description
1	Your login: HUGO.AB.MUELLER You are logged on! Password not saved [Save password]
2	[next]

The private start page then appears.

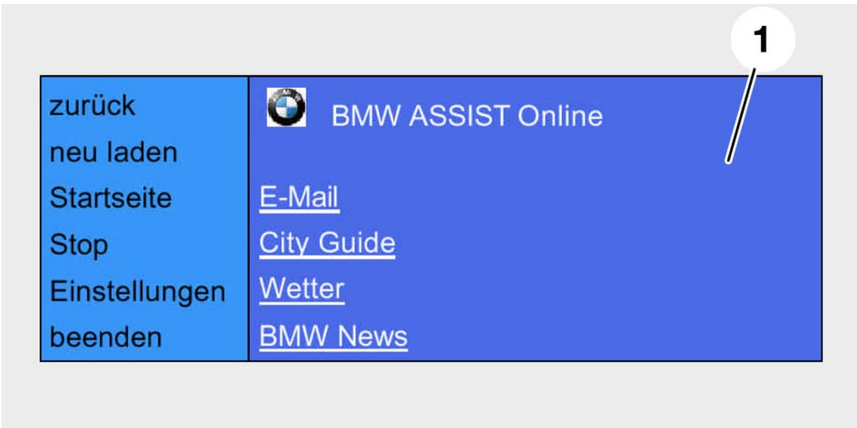


Fig. 74: Start page

Index	Description
1	BMW ASSIST Online E-mail City Guide Weather BMW News

b) Password is incorrect

If logon fails the portal generates the following message: Logon failed!

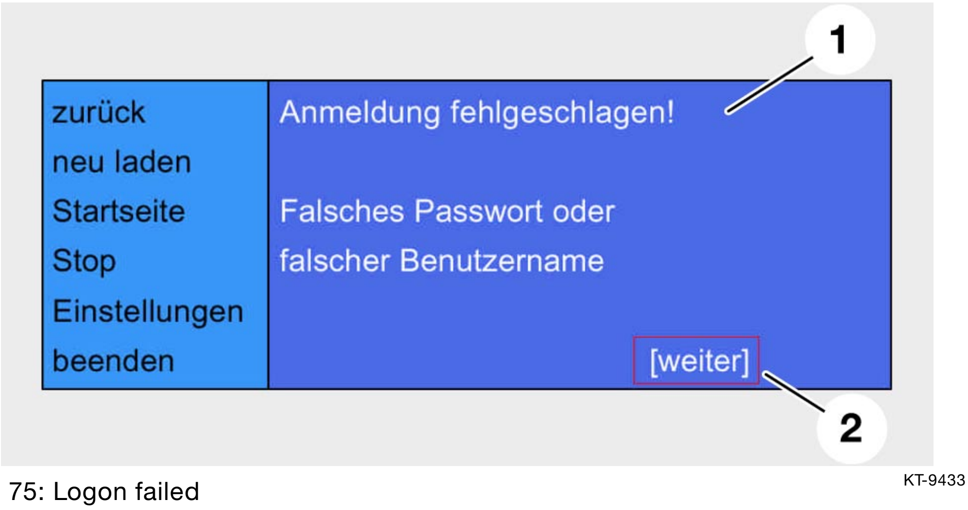


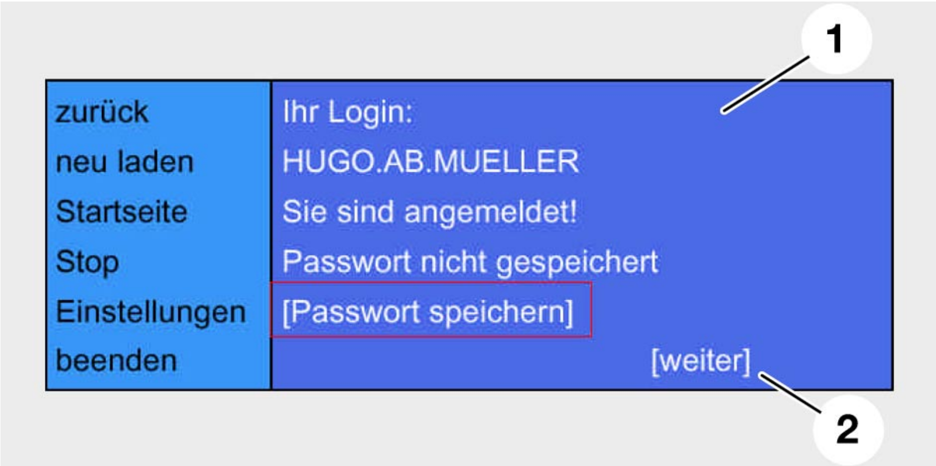
Fig. 75: Logon failed

Index	Description
1	Logon failed! Incorrect password or incorrect user name
2	[next]

The logon screen reappears when the user touches the "[next]" link.

2nd possibility

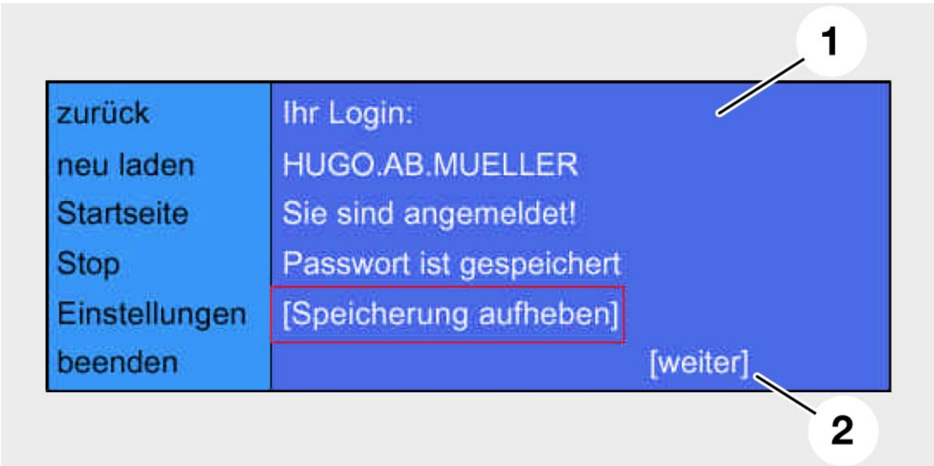
If the customer saved the password when he/she logged on at some previous point in time, it is not necessary to reenter the password. A message indicating that the password can be deleted from memory appears on the screen.



KT-9434

Fig. 76: Confirmation of logon, password already saved in memory

Index	Description
1	Your login: HUGO.AB.MUELLER You are logged on! Password not saved [Save password]
2	[next]



KT-9435

Fig. 77: Successful logon

Index	Description
1	Your login: HUGO.AB.MUELLER You are logged on! Password saved [Delete password from memory]
2	[next]

Logon as guest

Logging on as a guest is necessary if a vehicle is used by different people whose identities are not known in advance. Circumstances in which this applies include, for example:

- Company fleet cars
- Rental cars
- Etc.

Precondition:

The guest must be a customer (with another vehicle) and thus registered with the portal.

The guest sees a "Registration screen."

The guest selects the guest link and confirms that he/she wishes to register. He is then asked to enter his user name using the typewriter:

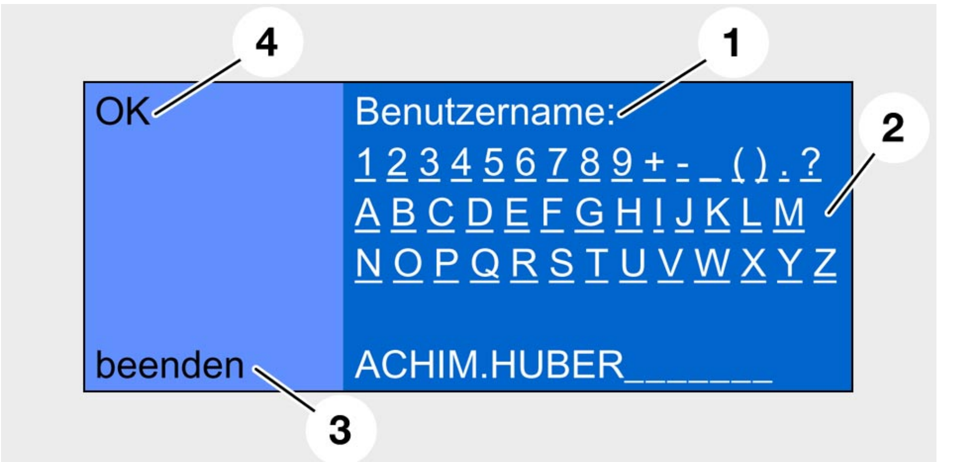


Fig. 78: Using the typewriter to enter the user name

Index	Description
1	User name:
2	Typewriter
3	exit
4	OK

The user presses OK to confirm the user name.
The login screen appears and the user is prompted to enter his/her password.

The user enters his/her password with the typewriter.

After OK has been pressed to confirm the entry, the portal checks whether the customer exists and verifies that the password as entered matches the user name.

a) If the outcome of these checks is affirmative, the portal sends the start page (homepage) of the user or customer to the vehicle.

b) The user in the hired car now uses the customer profile that he/she has available in his/her own car. In this way BMW customers have access to the full range of services in any car.

c) If the password and/or user name is incorrect, a message to this effect is issued and the logon procedure is aborted.
The registration screen reappears and the guest can try again.

Possible errors:

- User types in a user name that does not exist
    - The portal blocks the guest access if the name is entered wrongly on a second attempt

- User types in an incorrect password
    - Same response as in the customer login procedure

- Updating customer data

The configuration has to be updated if the car is deregistered or sold, the customer has purchased another car and wishes to use the Online services in the new car, or if the customer no longer desires access to the BMW ASSIST Online services.

The customer notifies the portal operator to the effect that he/she is a customer with a new vehicle or no longer wants access to the BMW ASSIST Online services. The portal operator must respond by updating the data record in the database.

1. Customer no longer wants access to the Online services:

   The portal deletes the data record from the database, which means that a customer is no longer stored in memory for the VIN of the vehicle in question.

2. Customer acquires a new car

   a. Customer has sold the original car and purchased a new car:
   The portal deletes the link to the original VIN but retains the data record and establishes a new link between this data record and the new VIN.

   b. Customer has another car along with the original car:
   The database with the VIN is retained in the database. The customer submits an application for BMW ASSIST Online services for his/her new vehicle. Instead of generating a new data record, the portal links the new VIN to the existing record.

3. Customer sells his/her vehicle, but has other vehicles registered for access to BMW ASSIST Online services:
   The existing data record is retained, only the link to the VIN of the vehicle sold by the customer is deleted.

The buyer of a BMW car must submit a new application in order to access the BMW ASSIST Online services. The portal creates a new data record in the database; this data record contains the VIN, the name of the customer, and the customer's telephone number.

The buyer of a BMW car wanting to log on to the portal for the first time must submit a new application.

A BMW ASSIST customer attempting to use the Online services from another vehicle is denied access. This is on account of the mismatch between the VIN registered for the customer and that of the vehicle.

Settings

The configuration has to be changed if, for example, the vehicle is sold. The "Settings" button is in the button bar on the left-hand side:

Fig. 79: The "Settings" button

Index	Description
1	Settings

All users have access to the "Settings" menu, so only the following changes are permitted:

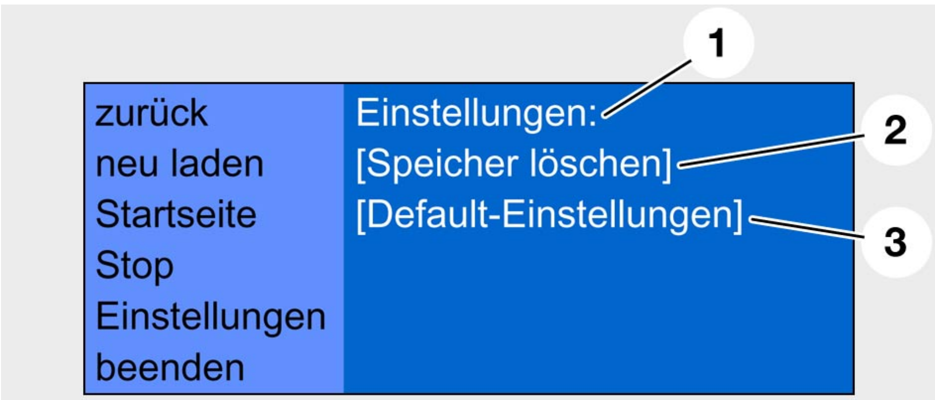


Fig. 80: The Settings menu

Index	Description
1	Settings:
2	[Clear memory]
3	[Default settings]

Explanations:

[Clear memory]	clears the entries in the on-board cache
[Default settings]	resets all settings to the factory defaults

All these actions are performed only locally, in other words in the vehicle: no connection is set up to the portal.

This, in turn, means that these actions can be performed after the customer has logged off.

In the current series release, the "Clear memory" and "Default settings" functions have been combined in the "Standard" button. When this function is triggered it clears the on-board cache and resets the parameters sent to the vehicle by the portal. This resets the parameters to their factory defaults. This can be helpful if problems arise with the browser or when new software is loaded onto the navigation computer. The function should always be triggered when the car is sold.

Using the services

The controller is the medium used to access the services.
The user turns the controller to select a service (e.g. [News]).

Fig. 81: Using the services KT-9439

Index	Description
1	News

The user presses in the controller to select the service; the first page of information is displayed on the screen.

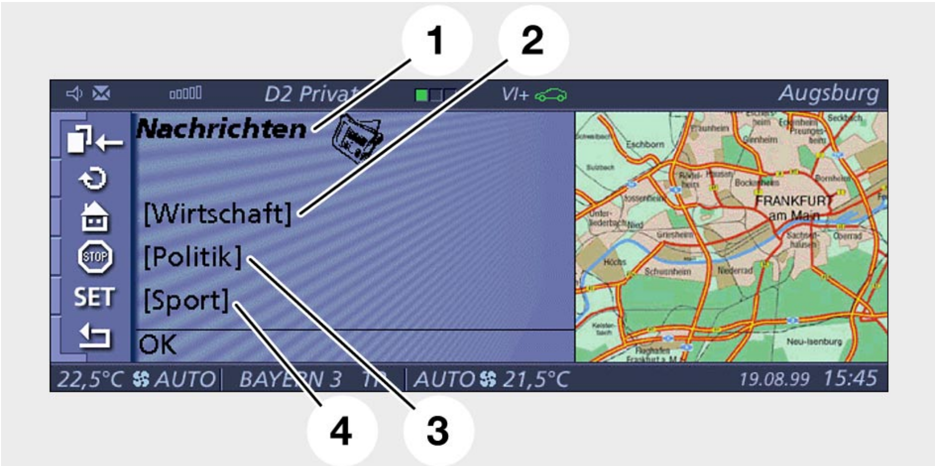


Fig. 82: The first page of the News service

KT-9440

Index	Description
1	News
2	[Economy]
3	[Politics]
4	[Sport]

The links in the bottom line are WAP-generated (by the simple simulation) and are not part of the planned Online services for the E65. They will be replaced by links for paging from one screen to the next.

A second page in the sequence of News pages could look like this (the last line is not repeated at the top of the next page):

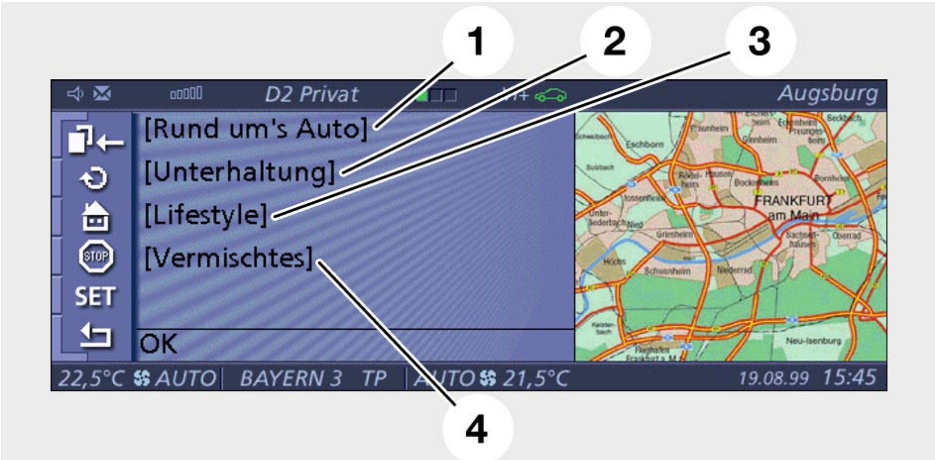


Fig. 83: The second page of the News service

Index	Description
1	[About cars]
2	[Entertainment]
3	[Lifestyle]
4	[Miscellaneous]

As this illustration shows, a complete information unit will not always fit on the screen, and in this case the user has to use the arrow buttons to page through the content.

Fig. 84: Screen showing part of the information available in the News service

Index	Description
1	Olympics: the torchbearers have just reached the Olympic stadium
2	Motocross: Joel Smets wins the 500 cc race
3	Page UP/Down buttons

WAP pages, too, can entail interaction with the user and request user inputs, as illustrated below:

KT-9424

Fig. 85: Data input is required

Index	Description
1	**Weather information** Please enter the post code:

KT-9423

Fig. 86: Typewriter and input line

Index	Description
1	Input line
2	Typewriter
3	Please enter the post code

KT-9443

Fig. 87: Information page

Index	Description
1	**Weather** Remaining clear and warm in the south while an area of low pressure will cause strong autumn storms in the north

At this time, data can be input only by means of the virtual typewriter. This also applies to the procedure for entering the password.

- Signing off

The browser is closed by the following actions:

- Clicking the Exit button on the button bar
- Ignition off
- As of 03/2002: Pressing the "On-hook" telephone button
- GSM connection interrupted
- Browser timeout (in other words no user action for 5 minutes)
- Emergency call

The browser "saves" the last page shown (except when the Exit button is clicked) and displays it when the next connection is set up.

- Notes for Service

Online services (like the telephone) use the mobile radio network via the air interface, so they are subject to the same physical laws and problems. For example:
Cell overload or poor network coverage cause the services to fail in just the same way as they interrupt or prevent calls on mobile phones.

Only one service can be used at a time, in other words telephone only or Online services only. Afterwards, the "old" connection has to be re-established.

Example: customer is online.
In this case the customer cannot receive incoming calls or place outgoing calls. The browser has to be actively closed before the telephone can be used.

The browser display is brighter than all other Control Display functions. This is to indicate to the customer that the information is made available through the portal and is not generated in the car. This means: if the display is faulty or the information is incorrect, the fault is not in the car and there is no need to replace parts unnecessarily.

Costs are incurred for the time that the customer is online. However, a timeout (approx. 5 minutes) is provided for. If no entries have been made within this period, the connection is interrupted automatically.
The green indicator on the Control Display that shows that a telephone connection has been set up flashes while an online data connection is in progress. This allows the customer to see at a glance whether or not a data connection has been established.

The function is always incorporated in the Navigation System 01, but has to be coded if it was not ordered as an option ex-works. Once coding has been completed the customer can register and log on as described above.

Telematics services, E65

- Introduction

The telematics services are an integral part of the Online Car Project of BMW.

The objective for the initial phase was to integrate the telematics functionality of the Mk-3 Navigation Computer (as at 09.00) and adapt them to the ergonomics of the E65.

Another objective was to make usage of the system as straightforward as possible.

- Functions

The groundwork of the telematics functionality in the E65 is the same as that of the E38.

In other words, all function of the Mk-3 navigation computer (as at 09/00) implemented by this data are available in the E65.

The following new functions, too, have been implemented.

- **Recommended deviation**
  The recommended deviation is at least as long as the route in question.

- **Automatic emergency call**
  Emergency services number dialled automatically when the telephone is not signed on.

- **Destinations list**
  Up to 20 destinations can be saved in the navigation system and picked up one after the other. If a new destination is requested from the provider it is entered at the top of the destinations list.

- Message list of traffic information

When V-Info Plus is called the traffic news is transmitted by the provider. When the individual messages are selected up to two pages of additional information are displayed.

Fig. 88: Message list of traffic information

Index	Description
1	Switch off
2	V-Info Plus
3	BMW
4	TMC transmitter
5	A9: Munich -> Nuremberg A8: Munich -> Augsburg A6: Nuremberg -> Heilbronn

Fig. 89: First page of additional information

KT-9383

Index	Description
1	**A9: Intersection Munich North -> Intersection Garching South** in 12 km traffic jam after accident, only one lane available Length: 10 km Delay: +20 min

Fig. 90: Second page of additional information

KT-9384

Index	Description
1	Caution! Fuel escaping! Do not throw hot or burning objects onto the road!

- **Emergency Service, local dealer, nearest dealer, hotline**

  Along with the "Emergency service" option, users can now select their local dealer or the nearest dealer or the hotline.

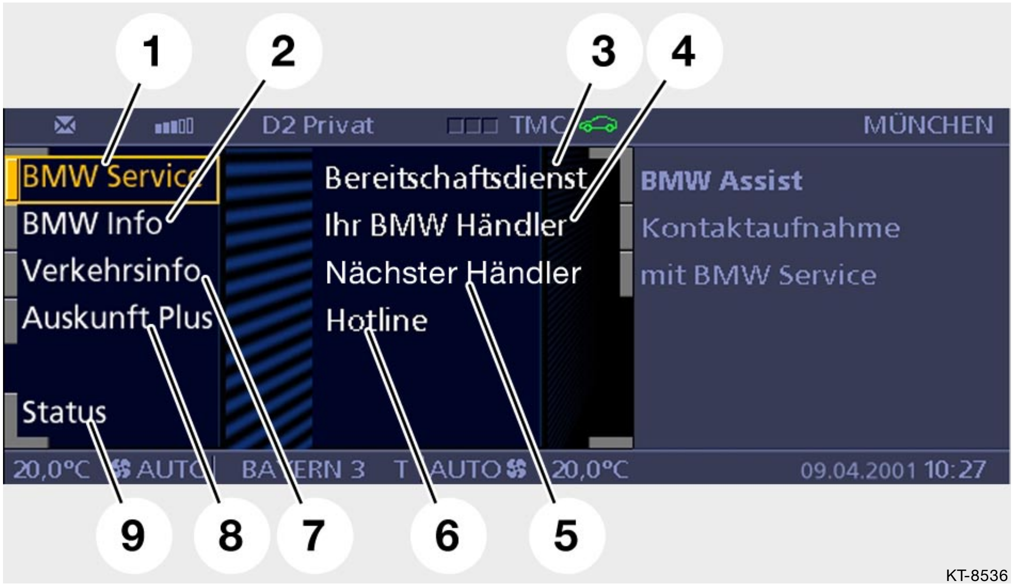


Fig. 91: BMW ASSIST menu

Index	Description	Index	Description
1	BMW Service	6	Hotline
2	BMW Information	7	Traffic Info
3	Emergency Service	8	Information Plus
4	Your BMW Dealer	9	Status
5	Nearest Dealer		

- "Emergency Service"

  The procedure for the "BMW Emergency Service" has not changed (key data is transmitted along with the position of the vehicle, if possible). The new feature is that the BMW Emergency Service is no longer tied to the navigation system (GPS data).

- "Local dealer"

  If the "Local dealer" button on the Control Display is pressed, the key data is read from the CAS (Car Access System). This data is sent to the telephone. An SMS (Short Message Service) is generated from this key data.
  This SMS is converted into an e-mail by the service provider and sent to the driver's local dealer.
  At the same time, the service provider sends another SMS back to the customer.
  Along with the confirmation, this SMS contains the telephone number of the customer's local dealer. A connection with this number is then set up (in much the same way as the Information service, for example). If the call is not accepted the dealer can call the customer back after reading the e-mail.

  The dealer generates the initial entry with the DISplus during the pre-delivery inspection.
  If the local dealer changes (e.g. because the vehicle is sold on or for some other reason), the user can access the BMW portal and change the address setting (my BMW).

- "Nearest dealer"

  The information defining the nearest BMW dealer is taken from the navigation CD as a navigation destination and displayed complete with telephone number on the Control Display. Data transfer between the vehicle and the service provider does not take place.
  This menu item does not appear if the car is not fitted with a navigation system.

- "Hotline"

  A connection is set up to a programmed telephone number.

- **No selection option for emergency call (in the Control Display)**
  An emergency call is triggered automatically in the event of an accident or manually by means of the "emergency call" button, so the ASSIST menu does not include an option for initiating an emergency call.

- **Configuration update**
  The automatic configuration update has a retry mechanism: 3 attempts with a two-minute wait between attempts.
  When roaming (network change) takes place, an automatic services configuration run is initiated. This is necessary on account of the differences between the services offered by individual providers.

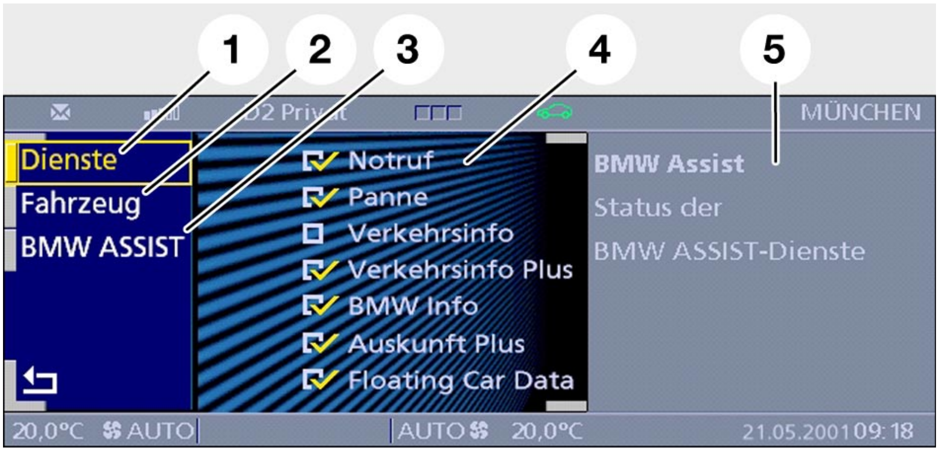


Fig. 92: Configuration update

KT-9382

Index	Description
1	Services
2	Vehicle
3	BMW Assist
4	List of services available
5	**BMW Assist** Status of the BMW ASSIST services

Voice processing system

- General

The voice processing system (SVS) of the E65 is an evolution of the know systems of the E38. If compared to the E38 voice processing system (up to 40 commands) the customer has now the possibility to choose among 400 commands.

The voice processing system is used in the MOST assembly as dialog means between the user and the whole system. It offers the improvement of the control and operability in the vehicle as well as the increase of comfort functions. The safety-related functions are not controlled. The voice input is always redundant in addition to buttons or Control Display functions.

The voice input system is an optional extra (SA) and is mounted only in association with the SA "telephone." The reason is the microphone of the hands-free conversation equipment, which is only mounted with the telephone equipment.

- Range of control functions

The language input system controls the following systems:

- Telephone
- Navigation (if installed)
- Radio
- Audio
- CD changer (if installed)
- Cassette player (if installed)
- Internet Browser
- Telematics
- Television (if installed)
- On-board computer
- User guide

The following wiring diagram shows the interconnections in the voice input system.

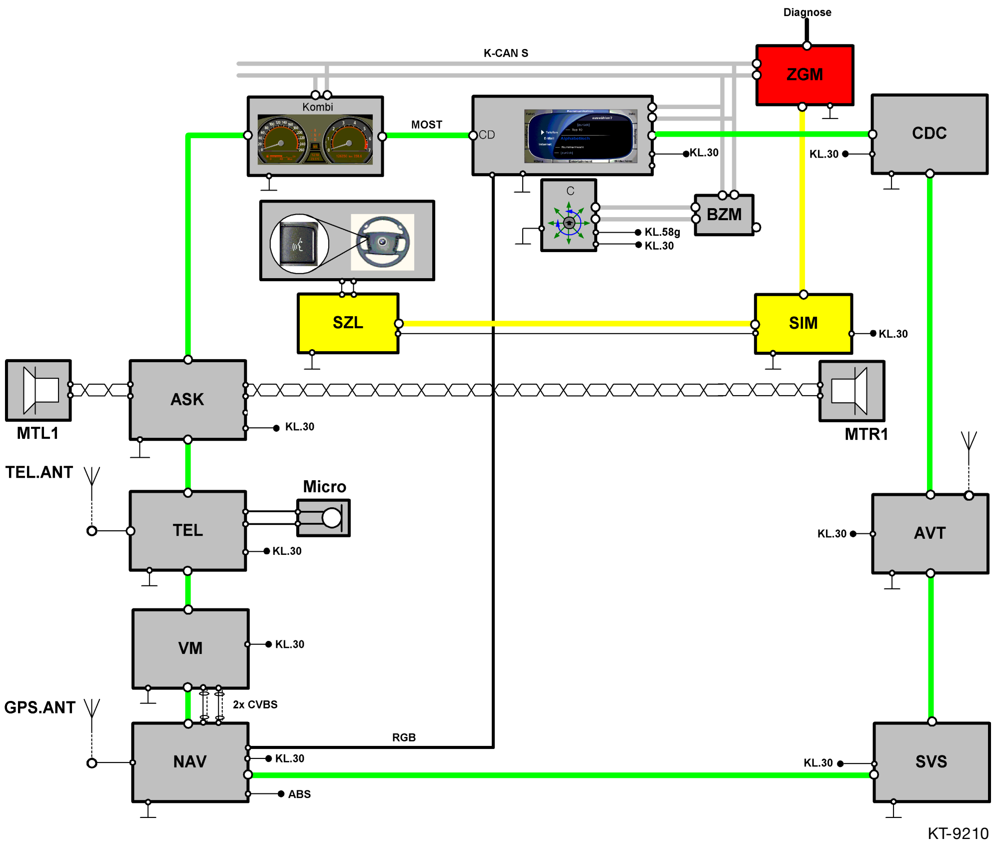


Fig. 93: Basic circuit of the voice processing system

Index	Description
ZGM	Central gateway module
Kombi	Instrument cluster
CD	Control Display
CDC	Audio system CD changer
BZM	Control panel module, centre console
C	Controller
SIM	Safety information module
SZL	Switching centre for steering column
PTT	Push-to-Talk button in the multifunction steering wheel
ASK	Audio System Controller
MTL 1	Mid-range speaker, front left
MTR 1	Mid-range speaker, front right
TEL.ANT	Telephone aerial
GPS.ANT	GPS aerial

TEL	Telephone
Micro	Microphone of the hands-free equipment
AVT	Aerial amplifier/tuner
VM	Video module
NAV	Navigation computer
SVS	Voice processing module
KL.30	Terminal 30 voltage supply, plus
KL.58g	Terminal 58g instrument lighting
ABS	ABS signal for wheel speed
RGB	Video signal cable
CVBS	Video signal lead
K-CAN S	System K-CAN
MOST	Media Orientated System Transport
byteflight	Safety bus
Diagnose	Diagnosis bus

- Functional description

Control philosophy

The voice processing system is a self-teaching system.
This means that each command has been learnt from approx. 100 persons with different dialects. The extreme dialects will be automatically discarded and a high recognition capacity will be attained.

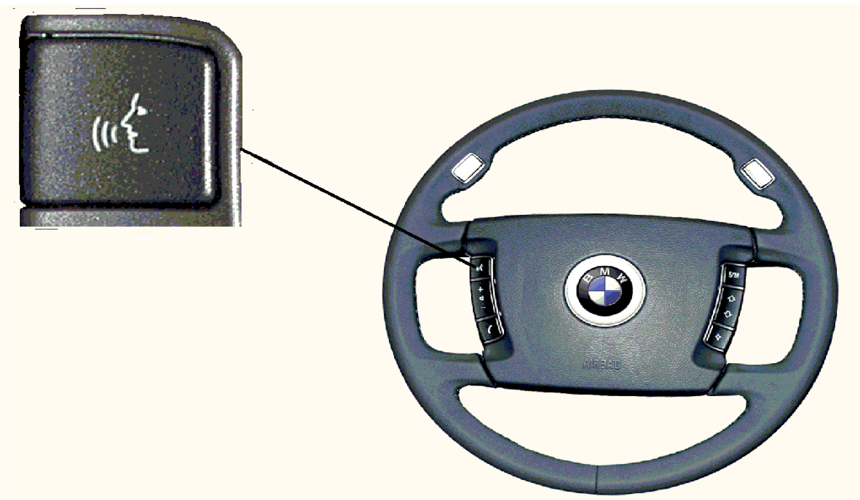
It is important that people speak as naturally as possible. They should not speak too loud nor whisper.

Operation

Start and abort, Push-to-talk button (PTT)

The system is started with the Push-to-talk button on the multi-function steering wheel of the vehicle. In SVS, the Control display requests a change in status. For this the Control display requests two audio channels to the Audio System Controller (ASK). One channel from the microphone to the SVS and one channel from the SVS to the front door-mounted loudspeakers, left and right. The SVS transmits to the Control display the execution of the change in status. The recognition dialog starts.

By pressing again the PTT button, the dialog can be ended at any time. Moreover, at any time the dialog can be ended by voice via the "Abort" command.



KT-9220

Fig. 94: PTT button on the steering wheel

Shut-off

While the dialog is active, the Control display can interrupt it (e.g. for a prioritary audio output). The SVS can stop the dialog, before the two audio connections are suppressed by the Control display.

The Control display informs the SVS, as soon as the recognition dialog can resume. In principle, the Control display can immediately interrupt the SVS. In this case, the Control display sends the status request "interrupt" to the SVS. There can then be no output on the audio channel. This process must however remain an exception.

General functions

By "general functions" we mean the functions which are part of all function groups and can always be recognised by the SVS, regardless of its dialog status.

The following commands are valid for all function groups:

- **Abort**

The current function is terminated.

- **SVS main menu**

The SVS is activated by pressing the PTT button and you reach then the main menu.

- **Help**

With this command, the help mode is activated. In the Help mode the operation is explained.

- **Options**

This command issues a list of all possible commands in the menu.

Full mode

The Full mode describes the complete dialog. Detailed task commands and forms for the commands to detect characterise the way the Full mode dialog is carried out. The Full mode is used to support the untrained operators and help them to perform the input required, e.g. "I must now *select a number*," "I would like to receive *Bayern 3*."

After a number of successful actions, the system invites the operator to switch to Quick mode.

Quick mode

The Quick mode is characterised by commands in their short format. The purpose of the Quick mode is to shorten the dialog for trained operators. e.g. "*Select number*," "*Play CD*."

When the SVS operated in Quick mode, if a command is rejected twice or if a Help function is needed, the SVS automatically switches back to Full mode.

Visual feedback

The E65 includes a visual feedback in the form of a text output on the instrument panel display. It displays in text form the SVS status message in the upper line of the instrument panel display. 20 characters are available. For the representation of the user input 18 characters are used in the lower line of the instrument panel display.

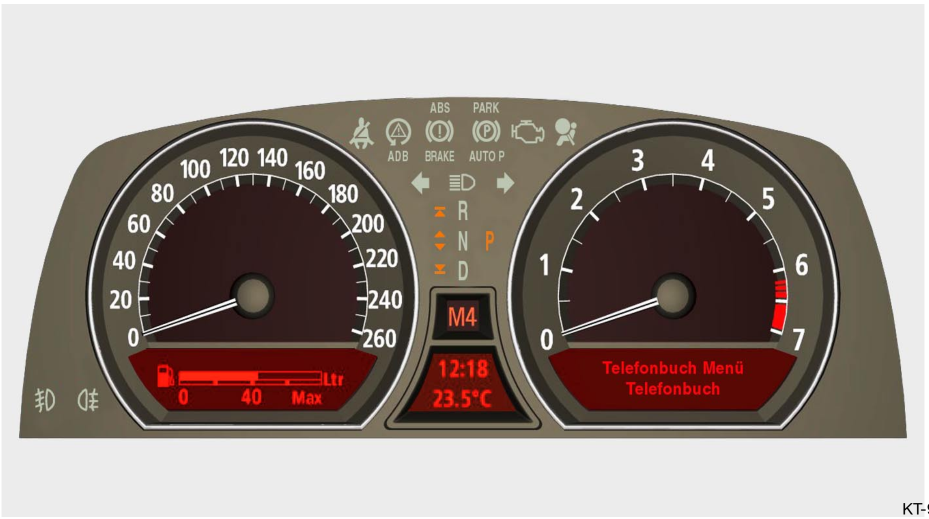


Fig. 95: Visual feedback in the Instrument cluster

Telephone

The following functions are available for voice control of the telephone operation:

- **PIN input**

The user has the possibility to enter the PIN number of the telephone card by voice.

- **Telephone switch on**

The telephone can be activated by voice.

- **Select number**

The user can enter any telephone number by voice. If the number is completely known, the selection process is started by a voice command.

- **Store names (internal to the SVS)**

The user can associate names and telephone numbers by a dialog and enter them into the telephone directory inside the SVS. The entries are made in sequence. The overall memory capacity is 100 names. When you store names entered by voice, make sure that a sample comparison is made with the names already stored in the directory.

- **Select names**

The names stored in the SVS internal directory can be issued by voice. The corresponding telephone number is then transmitted to the telephone and the connection is established.

- **Select**

- **Repeat selection**

This functions repeats the last number selected.

- **Display list of SMS / E-Mail**

With this function a list of all SMS received on the telephone is displayed on the On-board monitor.

- **Copy SIM telephone directory (internal to SVS)**

The SVS has the option to read the telephone directory of the SIM card inserted in the telephone. Then the single entries can be associated to the individual names by means of a dialog. This requires a comparison with the names already stored in the directory. The SVS takes care of synchronising and matching the data from different SIM cards. It is possible to handle and distinguish several SIM cards by their SIM-ID. The telephone always sends the latest data of the telephone directory in the current SIM card to the SVS.

- **Read telephone directory (SVS internal)**

The telephone directory inside the SVS is read sequentially. You can jump back and forth in the directory by voice commands. With a voice command the user can select the entry which has just been read from the directory. Then the telephone connection is established.

- **Delete a telephone directory entry (SVS internal)**

The user has the possibility to delete a single entry of the directory. Via a dialog, the user must enter the name corresponding to the entry to be deleted. For security reasons, the user must confirm it once before the entry is actually deleted.

- **Delete a telephone directory (SVS internal)**

The user has the possibility to delete all entries of the directory. For security reasons, the user must confirm it once before the entry is actually deleted.

Navigation

The following functions are available for voice control of the navigation system:

- **Store destination (SVS internal address list)**

The user can associate names and destinations by a dialog and enter them into the telephone directory inside the SVS. The entries are made in sequence. The overall memory capacity is 50 names. When you store names entered by voice, make sure that a sample comparison is made with the names already stored in the directory.

- **Select destination (SVS internal address list)**

The names stored in the SVS internal address directory can be issued by voice. The destination corresponding to the name is then transmitted by a voice command to the navigation system and the route guidance starts.

- **Destination input**

The destination input menu of the navigation system is activated.

- **Destination "Home"**

The navigation system takes the home address as the destination address and starts the route guidance.

- **Route guidance on/off**
- **Information on/off**
- **2-D Map**
- **Set scale**

With this function, you can change the scale of the map. By entering scale parameters, the user can define the scale directly. (100 m / 200 m / 500 m / 1 km / 2 km / 5 km / 10 km / 20 km / 50 km /100 km). He/she can also enlarge or reduce the scale step by step via voice commands.

- **Store position**
- **Display Point of interest (POI)**

By voice input, the following POIs can be displayed: Hotels / Service stations / Parkings at location / Destination.

- **Display new route**
- **Display last destinations**
- **Speedway selection**
- **By-road route selection**
- **Fastest route selection**

Radio

The following functions are available for voice control of the radio system:

- **Radio on / off**
- **Radio autostore**
- **AM/FM selection**
- **Frequency selection**
- **Traffic info on / off**
- **Radio station forward/back**

Audio

The following functions are available for voice control of the audio system:

- **Audio off**

CD changer

The following functions are available for voice control of the CD changer:

- **Track selection**

Cassette drive

The following functions are available for voice control of the cassette drive:

- **Cassette on / off**
- **Side change**

Television (TV)

The following functions are available for voice control of the television:

- **TV on / off**
- **Program forward / back**
- **Teletext setting**

Internet Browser

The following functions are available for voice control of the Internet Browser:

- **EBA on / off**
- **Call up BMW on-line/Internet**
- **Browser function Next page /Back**
- **Browser function Stop**
- **Browser function Home**
- **Browser function Reload**

Telematics services

The following functions are available for voice control of the Telematics services:

- **V-Info on / off**
- **BMW Emergency Service**
- **TMC on / off**
- **BMW Information**
- **BMW Info plus**
- **V-Info plus on / off**

On-board computer

The following functions are available for voice control of the board computer:

- **Show board computer**

Message block (SVS-internal)

The message block enables reception of voice messages. These are the functions available:

- **Receive messages**

You can receive several messages. These are received in sequence. The single messages are separated by a separator word or a signal. Overall, you have 5 minutes available to receive messages. 10 s before the memory is full, a visual warning is issued on the Instrument cluster display. The reception stops if you press the PTT button or after detecting a long pause.

- **Read messages**

You can listen to complete Message block or to individual messages. Via the function "forward" or "back" you can jump between messages. Between the single messages, a signal or separator word is output.

● Delete message

You can delete the complete Message block or individual messages. In particular, you can delete the message you just listened to. For security reasons, the user must confirm once before the message is actually deleted.

Glossary

Index	Description
AM	Amplitude modulation/Frequencies for long, medium, and short waves
ARD	Public work group of the Radio Authority of the Federal Republic of Germany
ASC	Automatic Stability Control
ASK	Audio System Controller
AVT	Antenna amplifier-Tuner
BIT2	Basic Interface Telephone II
Browser	Program for searching, retrieving and displaying on a computer screen data and documents on the Internet.
CAN	Control Area Network data bus system
CC	Compact Cassette/Audio Cassette
CD	Control Display/Display unit in the Comfort Area
CD	Compact Disc/Audio Disk
CDC	Compact Disk Changer/Audio CD changer
CVBS	Composite Video Burst Sync.
DAB	Digital Audio Broadcast/Digital radio reception
DECT	Digital Enhanced Cordless Telecommunications
DISplus	Diagnosis and Information System of 2nd generation
DSC	Dynamic Stability Control
DSP	Digital Sound Processor
ECE	European Commission for Europe, European standard
FABS	Colour image control and synchronisation signal, carrying colour, brightness and synchro pulses on a signal line (CVBS).
FBD	Remote control services
FM	Frequency Modulation = Very high frequency wave (VHF)
GAL	Speed-dependent volume regulation
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HF	High frequency
HTML	Hypertext Markup Language HTML is a programming language used to develop the web sites.
ISM	Industrial Science Medical Band
JBIT2	Japan Basis Interface Telephone 2
JBUF	Japan Mobile Phone Buffer, compensator for mobile phone aerials
Kombi	Instrument cluster
LA	Country-specific version

Index	Description
Links	Connections which make you jump from the current page to other pages or offers on the Internet.
LOGIC7	TOP HiFi amplifier
MD	Mini Disc/Mini audio disks
MoDiC	Mobile Diagnosis Computer
MOST	Media Oriented System Transport/Multimedia network
NAV	Navigation system
NF	Audio frequency
OCN	On Chip Navigation. The OCN-plug-in card in the navigation computer allows a quicker screen or map display.
Password	A password is a personal identification code
PDC	Park Distance Control
PDC Standard	Personal Digital Communication Standard Japanese standard similar to GSM
Pin	Personnel Identity Number / Secret number
Portal	Home page of a provider who offers various programs, e.g. T-online, YAHOO, etc.
PTT button	Push to talk button / Pushbutton to activate the voice processing system
RAS	Remote Access Service. A software which enables other users to select the BMW Portal.
RDC	Tyre inflation pressure control
Rendering Engine	Part of the browser, which defines the layout of the display.
RGB	Colour definition via the values for Red, Green and Blue. With 24-bit graphics, each colour component may have 256 levels.
RGBC	Colour definition via the values for Red, Green and Blue. With 24-bit graphics, each colour component may have 256 levels. Additionally, a fourth line is provided to transmit a separate synchronisation signal.
SA	Special version
SBDH	Cordless keypad handset
Server	Computer, which handles applications and documents so that they can be accessed by other computers.
SIM	Safety Information Module
SIM card	Subscriber Identity Module / Telephone identification card
S-Video	Analog video signal, which independently processes the brightness and the colours and provides a higher resolution and colour mapping.
SVS	Voice processing system
SWF3	Südwestfunk 3 (German broadcasting channel)
SZL	Switching centre for steering column

Index	Description
TEL	Telephone
TMC	Traffic Message Channel
TV	Television
VICS	Traffic information announcements in Japan
VM	Video module
WAP	Wireless Application Protocol. A technology standard, enabling the transmission of specially programmed Internet pages to a mobile terminal device (mobile phone).
WAP Cards	The WAP Card is the section to appear on the display. A WAP Card is a part of a WML page. (Deck)
WAP deck	The WAP deck is the complete WML page which is transmitted.
WAP mobile phone	Mobile phone equipped with a special Internet software to navigate on the Internet and display the retrieved pages.
WDCT	Worldwide Digital Cordless Telephone
WML	Wireless Markup Language is a programming language to display text on a mobile phone display.
ZDF	Second German TV channel
ZGM	Central gateway module